
Continual Learning With Participation Privacy: An Auditable Buffering-Aggregation Recipe

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Abstract

Modern federated and streaming learning systems often release intermediate models, so privacy must hold for the full trajectory under adaptive interaction. Motivated by participation privacy, we study single-edit neighboring user streams, where one insertion/deletion shifts all subsequent updates and defeats standard Hamming-neighbor continual-release analyses. We give an auditable modular recipe. A randomized buffering wrapper emits bins of size $[U, 2U]$, reducing single-edit streams to a Hamming-style per-bin update stream with explicit backlog/delay guarantees, where U is calibrated by the privacy parameters (ϵ, δ) . We then prove a certification theorem identifying when a non-adaptive Hamming-neighbor DP proof for a continual primitive lifts to adaptive inputs: the primitive must use fresh per-round randomness and have a stable one-round privacy profile under common adaptive context. Together, these ingredients yield trajectory-level (ϵ, δ) -DP for single-edit streams using standard primitives (e.g., tree prefix sums), with an explicit privacy-latency link via U .

1. Introduction

Modern federated and streaming learning systems release intermediate model snapshots throughout training, so an adversary may observe the entire trajectory $(w^{(t)})_{t \geq 0}$ rather than only a final model. A canonical instance is streaming stochastic gradient descent (SGD): at step t , a user contributes a private loss function f_t and the

learner updates using a gradient step based on the current model (Thakurta & Smith, 2013; Kairouz et al., 2021). Beyond protecting the data within each f_t , some deployments also require *participation privacy*: a user may wish to ensure *plausible deniability* of whether they participated at all. This motivates *single-edit* (edit-distance) neighboring streams, where two user-update streams are neighbors if one can be obtained from the other by inserting or deleting a single user event.

Continual release also makes the learning process intrinsically interactive: the released model $w^{(t)}$ influences what happens next, so the update stream can be adaptively generated in response to prior outputs. While classical continual-release primitives such as tree-based private prefix sums (Dwork et al., 2010a; Chan et al., 2011) were originally analyzed under statically chosen (Hamming-neighbor) streams, recent work formalizes differential privacy against such adaptive interaction (Jain et al., 2023). Our goal is to make this adaptive viewpoint compatible with the *single-edit* participation model above.

The difficulty is that Hamming-style neighboring streams are not the right abstraction for participation privacy on a stream. Most continual DP analyses change the value at a single time index t_0 while keeping all other positions aligned (Dwork, 2006). In contrast, a single insertion/deletion shifts the alignment of all subsequent positions, so an edit-neighbor pair can be far from Hamming-neighboring. This also undermines deterministic batching: partitioning the stream into fixed contiguous blocks can cause one edit to shift many later batch boundaries, so standard Hamming-neighbor continual DP guarantees do not directly transfer. Appendix G.1 gives an explicit boundary-shift attack on deterministic batching under edit adjacency, and we corroborate this failure empirically in our experiments.

To address this mismatch, we use a simple modular recipe: randomized buffering and an adaptive-safety certification for downstream continual primitives. We apply a *randomized buffering* wrapper (Chan et al., 2022; Zhou et al., 2023) that introduces random delay and releases user updates in *bins* of controlled size (e.g., in $[U, 2U]$); here U is a *privacy-implied systems cost*, calibrated by the target (ϵ, δ)

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and inducing delay. We then invoke standard continual DP primitives (e.g., tree/prefix-sum) to privately aggregate the resulting vector updates, and prove a *certification theorem* showing when a non-adaptive Hamming-neighbor privacy proof remains valid under feedback. A single edit can alter at most two adjacent emitted bins; adaptive group privacy therefore extends the certified distance-one Hamming guarantee to the resulting downstream prefix-sum inputs. The certification requires fresh per-round randomness together with a stable-context condition: common adaptive updates may change the released statistic, but they must not change the privacy cost of the unique Hamming discrepancy.¹

In the adaptive privacy game, the two candidate functions or bins are converted to updates using the same currently observed model within each branch; the two realized update streams need not remain Hamming-neighboring after their model trajectories diverge.

Contributions.

- **Edit-neighbor continual learning under feedback.** We formalize participation privacy via single-edit neighboring user-update streams in a continual learning setting with intermediate releases.
- **Modular pipeline with an explicit privacy to latency link.** We combine randomized buffering with standard continual DP primitives to obtain trajectory-level privacy for edit-neighbor streams; the target (ϵ, δ) determines the buffering level U , making induced delay explicit.
- **Certification theorem for adaptive safety of continual DP primitives.** We give a checkable certification condition under which a non-adaptive Hamming-neighbor privacy analysis carries over to the feedback/adaptive setting. The condition combines fresh per-round randomness with a stable-context requirement, and certifies safe reuse of canonical primitives such as tree/prefix-sum mechanisms.
- **Empirical validation (streaming SGD).** We report privacy–utility–latency tradeoffs under (ϵ, δ) -DP (fixing δ and sweeping ϵ), including delay distributions implied by $U(\epsilon, \delta)$; further instantiations are deferred to the appendix.

Results and organization. Our end-to-end guarantee is trajectory-level (ϵ, δ) -DP for *single-edit* neighboring participation streams under feedback, obtained by composing

¹The pre-conference version stated the certification condition using independent decomposability alone. However, we use the slightly stronger stable-context condition, formalized in Appendix E as quantitative decomposability. This refinement makes the statement technically precise and is satisfied by the certified cores of the continual-release primitives used in our pipeline.

randomized buffering with a continual-release DP aggregator (Theorem 3.1); the privacy target explicitly induces a buffering level $U(\epsilon, \delta)$ and hence delay/staleness.

A second ingredient is a reusable *certification* theorem showing that, for continual primitives satisfying our fresh-randomness and stable-context conditions, a standard non-adaptive privacy analysis lifts to the adaptive/feedback setting (Theorem 4.3). Section 2 formalizes the interactive model, edit adjacency, and metrics; Section 3 gives a high-level view of the pipeline and states the main theorem; Section 4 develops the two auditable ingredients (randomized buffering and certification) and combines them via modular composition; Section 5 instantiates the recipe with continual DP-SGD (tree/prefix-sum aggregation), and experiments appear in Section 6. Technical proofs and additional instantiations are deferred to the appendix.

1.1. Related Work

Continual-Release DP. Canonical continual-release primitives for streaming aggregates include tree-based private prefix sums (Dwork et al., 2010a; Chan et al., 2011) and closely related lower-triangular (matrix-style) mechanisms (Denisov et al., 2022). These mechanisms are typically analyzed for *Hamming-neighboring* update streams, where only one time index differs. In contrast, our participation model uses *single-edit* (insertion/deletion) neighboring streams, where alignment shifts after the edit; this gap is mild for static datasets (Birrell et al., 2024) but is fundamental for streams.

Sliding-window variants of private stream release have also been studied for window aggregate queries and graph streams (Cao et al., 2013; Upadhyay et al., 2021). These works target recent-window utility, whereas our setting focuses on single-edit participation privacy, where one insertion/deletion shifts all subsequent stream positions and requires an edit-to-Hamming shielding interface.

Adaptive Streams and Feedback. Continual learning is inherently interactive: released outputs can influence future updates. Recent work formalizes differential privacy against such adaptive interaction (e.g., left-or-right style games and verification viewpoints) (Denisov et al., 2022; Jain et al., 2023). Our *certification* result is complementary: it gives a checkable structural condition under which a standard non-adaptive Hamming-DP analysis of a continual primitive lifts directly to the adaptive setting. The condition is stronger than fresh prefix-causal randomness alone: it also requires that common adaptive context does not change the one-round privacy cost of the unique Hamming discrepancy.

Privacy-Induced Delayed Reactions. Cohen et al. (2024) give a conceptually related delayed-reaction phenomenon

for continual observation and online threshold queries. Their Appendix B delays threshold-type outputs on a directly observed online bit stream. Our delay has a different technical role: RandBin waits for future arrivals to form bins in $[U, 2U]$ so that a single insertion/deletion, which appears only in the neighboring-stream analysis, becomes a bounded Hamming-style perturbation while the full transcript, including timing and $\perp$ outputs, remains protected.

Randomized Buffering, RSC, and Obliviousness Linage. Randomized buffering has appeared as infrastructure in oblivious data structures and related privacy modularity work (Chan et al., 2022; Zhou et al., 2023). A closely related random-partition idea also appears in the Reorder-Slice-Compute (RSC) paradigm of Cohen et al. (2023), where random slicing is used to synchronize neighboring executions and avoid a one-edit domino effect. Our setting differs in that the continual streaming transcript exposes release timing itself: the adversary observes whether each step emits a real block or the dummy symbol $\perp$. Thus, in our use, RandBin is an interactive interface that converts single-edit participation streams into a sparse, Hamming-style per-bin update stream while also providing explicit backlog/delay guarantees calibrated by (ε, δ) .

Composition and Concurrency. Beyond classical (ε, δ) -DP (Dwork, 2006), alternative formalisms (e.g., Rényi divergence and tradeoff-based views) can streamline composition reasoning (Mironov, 2017; Dong et al., 2022; Vadhan & Zhang, 2023; Zhou et al., 2024), recovering familiar advanced composition bounds (Dwork et al., 2010b; Kairouz et al., 2015). Concurrent composition in interactive settings has also been studied, including adaptively chosen privacy parameters (Haney et al., 2023; Henzinger et al., 2026). These works typically assume each mechanism’s neighboring relation is defined on a single underlying (static or dynamic) database; in our pipeline, we additionally need to reason about *neighbor-preserving transformations* between components (edit-to-Hamming shielding), which motivates our modular composition viewpoint. Our modular composition theorem can be viewed as combining the refinement-pair reduction of NPDP to standard DP in (Zhou et al., 2024) with the interactive/concurrent composition framework of (Henzinger et al., 2026); for completeness and to match our power-function accounting, we give a self-contained proof in Appendix C.

Conflict of Interest Disclosure. The authors declare no financial or other substantive conflicts of interest relevant to this work.

2. Setting and Preliminaries

We fix the interactive streaming-learning model underlying continual release. The adversary observes both the released

model trajectory and release timing, and may influence future user events. We also define participation adjacency and the utility–latency metrics used throughout. Section 3 then summarizes our buffering–aggregation pipeline and main guarantee.

Streaming Learning Loop and Transcript. At each step $t = 1, 2, \dots$, an adaptive environment produces a *user event* f_t (e.g., a private loss or update). Starting from a public, data-independent model $w^{(0)} \in \mathbb{R}^d$, the server maintains models $w^{(t)}$ and may release intermediate snapshots. To capture both per-step and buffered releases, let $v_t \in \{0, 1\}$ be the observable timing bit: if $v_t = 1$, the server incorporates available buffered events, forms an update g_t , and publishes the new snapshot; if $v_t = 0$, it holds $w^{(t)} = w^{(t-1)}$. The adversary’s view through horizon T is

$$\text{tr}_{\leq T} := \left(w^{(0)}, (v_t, w^{(t)})_{t=1}^T \right).$$

Our privacy notion protects this joint transcript, including timing, with the same parameters for every horizon T .

Feedback/Adaptive Interaction. The environment, and hence the event stream, may depend on the past transcript: f_t may be an arbitrary randomized function of $\text{tr}_{< t}$. All guarantees quantify over such adaptive interaction. A *non-adaptive* analysis of a continual primitive instead assumes that its input stream is fixed independently of past releases.

Participation Privacy via Single-Edit Adjacency. For event streams $F = (f_1, f_2, \dots)$ and $F' = (f'_1, f'_2, \dots)$, write $F \sim_{\text{edit}} F'$ if one is obtained from the other by inserting or deleting one event, shifting all subsequent indices. Equivalently, for some t_0 , either

$$f'_t = f_t \quad (t < t_0), \quad f'_t = f_{t+1} \quad (t \geq t_0),$$

or the reverse relation holds. Thus the unit of privacy is one participation event.

Differential Privacy Under Adaptive Interaction. A fixed-stream DP inequality alone does not capture feedback. Formally, consider any adaptive paired environment $\mathcal{A}$ that, from the past observable view, generates two candidate event prefixes satisfying the prescribed edit-neighbor relation. Let $\text{View}_{b, \leq T}^{\mathcal{A}}$ denote its complete view when the mechanism receives candidate $b \in \{0, 1\}$. We require, for every $\mathcal{A}$, horizon T , and measurable set S ,

$$\Pr \left[\text{View}_{0, \leq T}^{\mathcal{A}} \in S \right] \leq e^\varepsilon \Pr \left[\text{View}_{1, \leq T}^{\mathcal{A}} \in S \right] + \delta,$$

and likewise with 0 and 1 exchanged. Fixed neighboring streams are the special case in which $\mathcal{A}$ ignores past releases. Appendix A formalizes this paired simulation and its equivalent power-function formulation.

Metrics. *Utility:* test accuracy or loss of the released model $w^{(T)}$. *Systems:* backlog Q_t , the number of pending events after time t , and inclusion delay

$$D(i) := \min\{t \geq i : f_i \text{ is incorporated by time } t\} - i,$$

measured in event arrivals.

Extension to Multiple Edits. Adaptive group privacy applies unchanged to paired simulations: an adaptive one-hop (ε, δ) guarantee yields, for distance at most k ,

$$\left(k\varepsilon, \delta \sum_{j=0}^{k-1} e^{j\varepsilon}\right)\text{-DP}$$

(Fact B.6). In particular, the two adjacent Hamming changes that a single edit may induce at our downstream interface are charged by this adaptive two-hop bound; the downstream one-hop parameters are calibrated accordingly in Section 3.

3. Overview

This section summarizes our buffering–aggregation pipeline (Fig. 1) and the resulting end-to-end guarantee: continual-release (ε, δ) -DP for single-edit participation streams under feedback, where the adversary observes the released trajectory (and timing). The required buffering level $U(\varepsilon, \delta)$ is privacy-implied and explicitly governs delay.

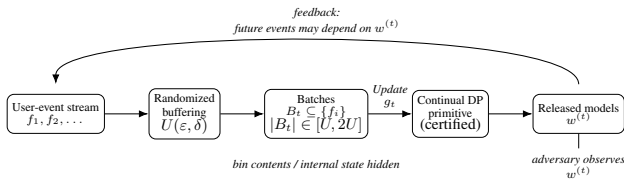


Figure 1. Pipeline overview. Randomized buffering (parameter $U(\varepsilon, \delta)$) converts a single-edit stream into bins that are consumed (in the sense defined in Appendix B) by a certified continual-release DP primitive, producing intermediate model releases. The adversary observes the released models; bin contents and internal states are hidden.

We next give the recipe and state the end-to-end guarantee.

Recipe (buffering $\rightarrow$ aggregation $\rightarrow$ releases). Given user-event stream (f_t) , at each step $t \geq 1$, buffering outputs B_t with $|B_t| \in [U, 2U]$ or $\perp$. Let $\tau(1) < \tau(2) < \dots$ be the (random) steps with $B_{\tau(k)} \neq \perp$.

1. **Buffer.** Run randomized buffering with $U(\varepsilon, \delta)$ to obtain $(B_t)_{t \geq 1}$.

2. **Update.** For each $k \geq 1$, form a clipped per-bin update $g_k := \frac{1}{|B_{\tau(k)}|} \sum_{f \in B_{\tau(k)}} \text{Clip}_G(\nabla f(w^{(\tau(k)-1)}))$, where $\text{Clip}_G(x) := x \cdot \min\left\{1, \frac{G}{\|x\|_2}\right\}$.
3. **Aggregate + release.** Feed $(g_k)_{k \geq 1}$ to a certified continual-release DP primitive and update only at bin times: output $w^{(\tau(k))}$ after processing g_k , and hold $w^{(t)} = w^{(\tau(k))}$ for $\tau(k) \leq t < \tau(k+1)$ (feedback allowed).

Theorem 3.1 states the resulting end-to-end privacy.

Theorem 3.1 (End-to-end continual-release DP for single-edit streams). *Run the above recipe with privacy target (ε, δ) , allocating $(\varepsilon_b, \delta_b)$ to buffering and $(\varepsilon_a, \delta_a)$ to the total downstream cost, where $\varepsilon_b + \varepsilon_a \leq \varepsilon, \delta_b + \delta_a \leq \delta$. Set*

$$U = \Theta\left(\frac{1}{\varepsilon_b} \log \frac{1}{\delta_b}\right), \quad \hat{\varepsilon}_a = \frac{\varepsilon_a}{2}, \quad \hat{\delta}_a = \frac{\delta_a}{1 + e^{\varepsilon_a/2}}.$$

Plug in any certified continual-release DP primitive that is adaptively $(\hat{\varepsilon}_a, \hat{\delta}_a)$ -DP for ordinary distance-one Hamming-neighboring update streams. Then the released model trajectory $(w^{(t)})_{t \geq 0}$ is (ε, δ) -DP with respect to single-edit neighboring event streams, even when future events are generated adaptively from past releases (feedback).

Proof idea. Randomized buffering turns a single insertion/deletion into at most two adjacent Hamming changes in the emitted update stream. Certification gives adaptive distance-one privacy, and adaptive group privacy (Fact B.6) charges the two changes within the allocated $(\varepsilon_a, \delta_a)$ budget. Modular composition with buffering then gives the claimed guarantee (Appendix C).

Certification preview (adaptive safety of continual primitives).

We use a reusable certification for feedback. Writing x_t for the per-round input and $x_{1:t}$ for its prefix, the certified core must use fresh independent randomness at each round and satisfy a stable-context condition: common adaptive inputs may change the released statistic, but they must not change the one-round privacy cost of a distance-one Hamming discrepancy. Under these conditions, a standard non-adaptive (ε, δ) -DP analysis lifts to the feedback setting. Section 4.2 gives the main-text checklist and theorem, while Appendix E formalizes the stable-context condition as quantitative decomposability. The certification covers tree/prefix-sum mechanisms and more general lower-triangular linear mechanisms.

SGD instantiation and what we measure. We instantiate the pipeline with streaming SGD: each emitted bin $B_{\tau(k)}$ triggers one clipped mini-batch gradient update, privatized by a continual-release DP aggregator (tree/prefix-sum in our experiments). The target (ε, δ) fixes $U(\varepsilon, \delta)$,

yielding random batch sizes in $[U, 2U]$ and a backlog bound of order $O(\frac{1}{\varepsilon_b} \log t (\log t + \log \frac{1}{\delta_b}))$ (Lemma 4.2). We report accuracy versus (ε, δ) together with empirical delay distributions implied by $U(\varepsilon, \delta)$. The theoretical convergence of various SGD variants under DP noise has already been extensively analyzed; we refer the reader to standard works (Jain et al., 2012; Song et al., 2013; Thakurta & Smith, 2013; Kairouz et al., 2021; Denisov et al., 2022).

Deferred details. Sections 4.1 and 4.2 formalize buffering and certification; the end-to-end proof of Theorem 3.1 (via modular composition) and additional instantiations/experiments appear in the appendix.

4. Auditable Ingredients: Randomized Buffering and Adaptive-Safe Continual-Release DP

This section isolates the two interfaces used in Theorem 3.1. Randomized buffering (RandBin) supplies an edit-to-Hamming reduction and an explicit backlog guarantee (Section 4.1), while Theorem 4.3 certifies when a non-adaptive Hamming-DP continual primitive remains private under feedback. Section 4.3 composes these interfaces.

4.1. Randomized Buffering Wrapper

RandBin Interface. Given user events $f_1, f_2, \dots$, at each step $t \geq 1$, RandBin outputs either a bin B_t of pending events or $\perp$ (no emission). Let $\tau(1) < \tau(2) < \dots$ denote the random emission times for which $B_{\tau(k)} \neq \perp$. When $B_t = \perp$, nothing is sent downstream and the released model remains unchanged. We use the following interface properties:

- (P1) **Bin size.** Whenever $B_t \neq \perp$, $|B_t| \in [U, 2U]$.
- (P2) **Backlog control.** The number Q_t of pending events remains small for every t , yielding explicit inclusion-delay guarantees.
- (P3) **Edit-to-Hamming shielding.** A single insertion/deletion induces coupled emitted-bin/update interface streams at Hamming distance at most two.

Input-Identifiability Convention. Lemma 4.1 uses the standard convention that events are identifiable, e.g., by unique IDs or metadata; anonymous duplicate events require an additional disambiguation assumption that is discussed in Appendix D.

Backlog and Delay. The buffering level $U(\varepsilon_b, \delta_b)$ determines how aggressively RandBin batches the stream. We use the backlog Q_t and inclusion delay $D(i)$ defined in Section 2; because events are emitted from a FIFO buffer, bounds on Q_t yield corresponding delay bounds.

Algorithm 1 RandBin (Interface-Level Skeleton)

```

1: Input:  $(\varepsilon_b, \delta_b)$ ; stream  $(f_t)_{t \geq 1}$ 
2: Output:  $B_t \in \{\perp\} \cup \{\text{bins of size in } [U, 2U]\}$ 
3: Set  $U = \Theta(\varepsilon_b^{-1} \log(1/\delta_b))$  and a distribution  $\mathcal{D}$  supported on  $[U, 2U]$ 
4: Maintain a FIFO buffer  $\text{buf}$  and a private randomized emission scheduler
5: for  $t = 1, 2, \dots$  do
6:   Append  $f_t$  to  $\text{buf}$ 
7:   if there is no scheduled emission then
8:     Output  $\perp$ 
9:   else
10:    Sample  $C \in [U, 2U]$  from  $\mathcal{D}$ 
11:    Pop  $C$  items from  $\text{buf}$  as  $B_t$ 
12:    Output  $B_t$ ; refresh scheduler
13:   end if
14: end for
    
```

Lemma 4.1 (Edit-to-Hamming Wrapper (NPDP Contract)). *For any single-edit neighboring input streams, RandBin admits a neighbor-preserving paired simulation (with refinement) in which the coupled emitted-bin/update interface streams have Hamming distance at most two. If two concrete-bin positions differ, they are adjacent.*

Intuition: randomized boundaries prevent one insertion/deletion from cascading into many downstream bin shifts; the paired simulation formalizes this stability under interaction.

Lemma 4.2 (Backlog Bound (Privacy-Implied Delay)). *With $U = \Theta(\varepsilon_b^{-1} \log(1/\delta_b))$, for every time t the backlog satisfies $Q_t = O(\frac{1}{\varepsilon_b} \log t \cdot (\log t + \log \frac{1}{\delta_b}))$.*

This bound is obtained by implementing RandBin’s private emission scheduler via the streaming DP prefix-sum mechanism (Theorem F.2) and translating its additive-error guarantee into a worst-case queue bound. As a consequence, inclusion delay is controlled at the same order (up to constants) for events arriving by time t .

Intuition: the noisy scheduling forces sufficiently frequent emissions to prevent sustained queue growth, while keeping bin sizes within $[U, 2U]$.

On the Horizon Dependence. The $\log t$ dependence in Lemma 4.2 is not merely algebraic. Our emission scheduler is a continual private prefix-sum mechanism whose additive error translates directly into backlog. On a unit-rate stream, if M_t denotes cumulative emissions by time t , then $Q_t = t - M_t$. Hence, within this scheduler family, improving backlog requires improving continual private counting error. Known lower bounds suggest that logarithmic horizon dependence is intrinsic to this proof route (Dwork et al., 2010a; Cohen et al., 2024), though we do not claim a minimax lower bound over all NPDP wrappers.

Scope of the Approximate-DP Regime. The probability-one finite-delay guarantee in Lemma 4.2 relies on the approximate-DP implementation of RandBin: its proof uses bounded-support buffering noise and an always-valid prefix-sum error bound. RandBin-style shift masking under pure DP requires unbounded-support noise and thus cannot provide the same guarantee. Appendix I discusses this limitation further.

Positioning. Randomized buffering is often used in oblivious data structures to hide access patterns (Chan et al., 2022; Zhou et al., 2023). Our emphasis differs: RandBin is an *infrastructure interface* that (i) converts single-edit participation streams into sparse, Hamming-style per-bin update streams and (ii) provides explicit back-log guarantees. Once Lemmas 4.1–4.2 hold, any continual primitive certified in Section 4.2 can be reused downstream. Full pseudocode, the NPDP security game (paired simulation and refinement), and sharper tail bounds appear in Appendix D.

4.2. Certifying Feedback-Safe Continual-Release Primitives

Checklist 1 (Certification Conditions). A continual-release primitive $\mathcal{M}$ passes certification if its core release process satisfies the following conditions:

1. **Prefix dependence:** the core release at time t depends only on the input prefix $x_{1:t}$, not on future inputs.
2. **Fresh randomness:** the core release at time t uses randomness R_t that is independent of the randomness used in all other rounds.
3. **Deterministic release map:** given $(R_t, x_{1:t})$, the core release $y_t = \mathcal{M}_t(R_t, x_{1:t})$ is deterministic.
4. **Stable one-round privacy cost:** if two prefixes differ in one update, then common aggregation context in the two prefixes does not change the privacy cost of that discrepancy. The appendix formalizes this condition as quantitative decomposability in Definition E.3.
5. **Post-processing only afterward:** after the certified core releases are produced, the mechanism may output any function of the released transcript; this is post-processing and does not weaken the privacy guarantee.

Consequence: any finite-horizon non-adaptive (ϵ, δ) -DP proof for Hamming-neighboring update streams lifts to adaptive interaction.

Why Certification? Our pipeline treats the continual-release DP primitive as a plug-in component. To make this modularity auditable, we need a criterion under which a standard non-adaptive Hamming-neighbor DP analysis remains valid when per-round updates are chosen adaptively from past releases. Fresh independent randomness is necessary but not sufficient; Appendix E.1 gives a two-round counterexample. Our criterion therefore adds quantitative decomposability, preventing adaptive common context from changing the one-round leakage profile.

Certified Decomposability. Write x_t for the input update at round t and $x_{1:t} := (x_1, \dots, x_t)$. The certification condition has two parts. First, the mechanism must use fresh independent randomness at each round: the release at time t is generated as

$$y_t = \mathcal{M}_t(R_t, x_{1:t}),$$

where R_t is independent of the randomness used at other rounds and $\mathcal{M}_t$ is deterministic given $(R_t, x_{1:t})$.

Second, the privacy cost of a one-round release must be stable under common context. Informally, if two input histories differ in one update, then the effect of this discrepancy on the round- t release is determined only by the part of the round- t aggregate that contains the changed update. Common updates before or after the discrepancy may change the released statistic, but they are shared by both neighboring histories and therefore should not change the privacy cost of the discrepancy itself.

The formal appendix condition, called quantitative decomposability, rules out this behavior by requiring the one-round privacy profile to depend only on the distance between the relevant aggregates after common aggregation context is removed.

Importantly, after producing the output $y_{1:t}$ at time t , the mechanism may apply *arbitrary* post-processing to $(y_1, \dots, y_t)$ and output any derived value at time t ; by closure under post-processing, this does not weaken the privacy guarantee proved for the underlying releases.

Certification Theorem. The next theorem formalizes the consequence of Checklist 1 and serves as an audit tool for reusing continual DP primitives inside interactive learning loops. The formal appendix statement is given in terms of power functions; the version below states the resulting (ϵ, δ) -DP guarantee.

Theorem 4.3 (Certification: Non-Adaptive DP Implies Feedback-Safe DP). *Consider Hamming-style neighboring update streams, where two equal-length streams differ in at most one position. Let $\mathcal{M}$ be a continual primitive satisfying Checklist 1. If $\mathcal{M}$ is non-adaptively (ϵ, δ) -DP for*

Hamming-neighboring update streams for every finite horizon, then $\mathcal{M}$ is also (ε, δ) -DP against adaptive interaction for every finite horizon, where each x_t may be chosen as an arbitrary function of past releases.

Proof Sketch. (Full proof in Appendix E) The proof works in the paired-simulation game and proceeds by induction on the horizon. If the first input pair already contains the unique Hamming discrepancy, then all later paired inputs must be equal componentwise. Checklist 1 ensures that these later common inputs only add common context and do not change the one-round privacy cost of the discrepancy; the per-round privacy losses then tensorize exactly. If the first input pair is equal, we condition on the common first release, fix the first input as common context, and apply the induction hypothesis to the shifted mechanism on the remaining rounds. Fresh independent randomness is used to separate the per-round contributions, while the stable-context condition prevents adaptive common inputs from amplifying the leakage of the unique discrepancy.

Canonical Examples. For the tree mechanism, take the certified core to be the newly revealed noisy tree-node values. Each uses fresh noise, and only nodes whose intervals contain the changed update have different aggregates; published prefix sums are post-processing of these values. For a lower-triangular mechanism, take as the core

$$z_t = \langle b_t, x_{1:t} \rangle + \eta_t, \quad \text{supp}(b_t) \subseteq \{1, \dots, t\},$$

with fresh independent η_t . Common updates contribute equally to the two neighboring executions, and any output $y_t = \sum_{s \leq t} A_{t,s} z_s$ is post-processing. Thus both mechanisms pass Checklist 1; details appear in Appendix F.1.

Scope Boundary. Theorem 4.3 is a sufficient certification condition, not a characterization of adaptive privacy. Fresh randomness alone is not enough; Appendix E.1 gives a two-round separation. The missing ingredient is the stable-context condition in Checklist 1, which prevents adaptive common inputs from changing the leakage profile of an earlier Hamming discrepancy. Conversely, failure of Checklist 1 does not imply that adaptive privacy is false; it means that a separate interaction-aware proof is needed, as for RandBin in Appendix D.

4.3. Putting the Ingredients Together

We view RandBin as producing a visible buffering transcript and a hidden per-bin update stream consumed by the downstream continual primitive. By Lemma 4.1, under a single input edit the two coupled interface streams have Hamming distance at most two.

Let the downstream primitive pass Checklist 1 and be non-adaptively $(\widehat{\varepsilon}_a, \widehat{\delta}_a)$ -DP for ordinary distance-one Hamming neighbors. Theorem 4.3 gives the same guarantee under adaptive interaction, and adaptive group privacy (Fact B.6) gives the distance-two guarantee $(2\widehat{\varepsilon}_a, (1 + e^{\widehat{\varepsilon}_a})\widehat{\delta}_a)$. Accordingly, we set $\widehat{\varepsilon}_a = \frac{\varepsilon_a}{2}$, $\widehat{\delta}_a = \frac{\delta_a}{1 + e^{\varepsilon_a/2}}$, so the total downstream cost is at most $(\varepsilon_a, \delta_a)$.

Finally, composing RandBin’s $(\varepsilon_b, \delta_b)$ -NPDP guarantee with this downstream guarantee using the modular composition theorem in Appendix C yields $(\varepsilon_b + \varepsilon_a, \delta_b + \delta_a)$ -DP for single-edit neighboring event streams, even under adaptive interaction. This proves Theorem 3.1.

Takeaway. Once the wrapper and certification conditions hold, the downstream primitive can be reused after calibrating its distance-one budget for the two-hop interface.

5. Instantiation: Continual DP-SGD Pipeline

This section instantiates the buffering–aggregation recipe (Section 4) with streaming SGD (Algorithm 2). Each arriving user event f_t is buffered by RandBin. When a bin is emitted, we form a clipped mini-batch gradient and pass a stepsize-scaled update to a continual-release DP aggregator. In our experiments, the aggregator PrivStreamSum is either (i) the standard tree-based prefix-sum mechanism (Chan et al., 2010), or (ii) the lower-triangular matrix factorization prefix-sum mechanism (Fichtenberger et al., 2023). Their formal DP and error/consistency guarantees are summarized in Appendix F. As in Fig. 1, the adversary observes the released models and release timing (i.e., whether $B_t = \perp$ at each step), while bin contents and internal states are hidden.

Fixed vs. Tuned Parameters. The privacy allocation determines the buffering level $U(\varepsilon_b, \delta_b)$ and reserves $(\varepsilon_a, \delta_a)$ for the total downstream cost. Since a single edit can alter at most two adjacent update positions, PrivStreamSum is run with the one-hop parameters

$$\widehat{\varepsilon}_a = \frac{\varepsilon_a}{2}, \quad \widehat{\delta}_a = \frac{\delta_a}{1 + e^{\varepsilon_a/2}}.$$

Given a learning task, the clipping norm G and stepsizes (η_k) are tuned for utility; once chosen, they determine the conservative one-hop update sensitivity

$$\Delta_u = 2G \max_{1 \leq k \leq T_{\max}} \eta_k.$$

The release schedule is induced by RandBin, with $|B_t| \in [U, 2U]$ whenever $B_t \neq \perp$. The plug-in aggregator must be non-adaptively $(\widehat{\varepsilon}_a, \widehat{\delta}_a)$ -DP for ordinary distance-one Hamming neighbors at sensitivity Δ_u and pass Checklist 1.

Algorithm 2 Continual DP-SGD Under Single-Edit User Streams

```

1: Input: Privacy target  $(\varepsilon, \delta)$ ; public horizon  $T_{\max}$ , i.e., an
   upper bound on both the number of time indices and emitted
   bins; allocation  $(\varepsilon_b, \delta_b)$  to buffering and  $(\varepsilon_a, \delta_a)$  to the total
   downstream cost, with  $\varepsilon_b + \varepsilon_a \leq \varepsilon$  and  $\delta_b + \delta_a \leq \delta$ ; initial
   model  $w^{(0)}$ ; clip norm  $G$ ; stepsizes  $(\eta_k)_{k \geq 1}$ 
2: Output: Released trajectory  $(w^{(t)})_{t=0}^{T_{\max}}$  and release timing
    $(\mathbf{1}\{B_t \neq \perp\})_{t=1}^{T_{\max}}$ 
3: Set  $U = \Theta(\varepsilon_b^{-1} \log(1/\delta_b))$  and initialize RandBin with level
    $U$ 
4: Set  $\hat{\varepsilon}_a \leftarrow \varepsilon_a/2$ ,  $\hat{\delta}_a \leftarrow \delta_a/(1 + e^{\varepsilon_a/2})$ , and  $\Delta_u \leftarrow$ 
    $2G \max_{1 \leq j \leq T_{\max}} \eta_j$ 
5: Initialize PrivStreamSum with one-hop budget  $(\hat{\varepsilon}_a, \hat{\delta}_a)$  and
    $\ell_2$ -sensitivity  $\Delta_u$ 
6:  $k \leftarrow 0$ 
7: for  $t = 1, 2, \dots, T_{\max}$  do
8:   Receive user event  $f_t$ 
9:    $B_t \leftarrow \text{RandBin}(f_t)$ 
10:  if  $B_t = \perp$  then
11:     $w^{(t)} \leftarrow w^{(t-1)}$ 
12:  else
13:     $k \leftarrow k + 1$ 
14:     $g_k \leftarrow \frac{1}{|B_t|} \sum_{f \in B_t} \text{Clip}_G(\nabla f(w^{(t-1)}))$ 
15:     $u_k \leftarrow \eta_k g_k$   $\triangleright$  stepsize-scaled update
16:     $S_k \leftarrow \text{PrivStreamSum}(u_k)$   $\triangleright$  private prefix sum
17:     $w^{(t)} \leftarrow w^{(0)} - S_k$   $\triangleright$  post-processing of  $S_k$ 
18:  end if
19: end for

```

Theorem 4.3 lifts this guarantee to adaptive interaction, and adaptive group privacy (Fact B.6) bounds the resulting two-hop cost by $(\varepsilon_a, \delta_a)$.

Privacy-Budget Split. A symmetric 50/50 split between buffering and the total downstream cost is a simple default. More generally, any fixed split $\varepsilon_b = c\varepsilon$ and $\varepsilon_a = (1 - c)\varepsilon$, with $c \in (0, 1)$, changes the asymptotic guarantees only by constant factors, while assigning constant fractions of δ changes $\log(1/\delta_b)$ only by an additive $O(1)$ term. The best finite-sample split can be selected by sweeping c on public or separately privatized validation data and measuring the latency–accuracy tradeoff.

Mini-Batch View. When a bin is emitted, the gradient estimate is the clipped mini-batch average

$$\frac{1}{|B_t|} \sum_{f \in B_t} \text{Clip}_G(\nabla f(w)).$$

Thus $\|g_k\|_2 \leq G$. Moreover, RandBin enforces $|B_t| \in [U, 2U]$, so the batch size varies by at most a factor of 2.

Implementation Notes and Measured Latency. The server maintains (a) the current released model $w^{(t)}$, (b) the FIFO buffer state of RandBin, and (c) the internal state of

PrivStreamSum (e.g., its noisy tree or factorization state). The model is updated only at emission times and is held fixed otherwise, yielding a piecewise-constant released trajectory. To quantify privacy-implied latency, we log the backlog Q_t of pending events after time t . For an event f_i , its inclusion delay is

$$D(i) := \min\{t \geq i : f_i \in B_t\} - i,$$

which can be computed from the same FIFO queue evolution underlying (Q_t) ; we report summary statistics and empirical CDFs in the experiments.

Other Optimizers. The same template applies to streaming optimizers whose per-bin updates can be encoded as a clipped, sensitivity-bounded stream for a certified continual-release aggregator: replace the definitions of g_k and u_k while keeping RandBin as the edit-to-Hamming wrapper and PrivStreamSum as the aggregation primitive. Variants with additional state, such as momentum or adaptive preconditioning, require a separate audit of Theorem 4.3’s conditions; falling outside the theorem does not by itself imply failure of adaptive privacy. Optimizer-specific privacy-loss saturation results are complementary to our backlog analysis; see Appendix I.

6. Applications and Experiments

We report experiments to demonstrate the framework.² The main text focuses on deep networks trained with SGD under different aggregation mechanisms. In Appendix G.1, we give an attack showing that a Hamming-style continual release mechanism (e.g., tree-based prefix sums) is not adaptive to single-edit streams; in Appendix G.2, we validate the delay phenomenon introduced in Section 4; in Appendix G.4, we give a synthetic ADMM study that fits our setting.

6.1. Deep-Network Experiments

We evaluate Algorithm 2 in a deep-network setting to study the utility impact of the randomized buffering wrapper RandBin under single-edit user streams. We use MNIST, CIFAR-10, and EMNIST ByMerge, where simple networks achieve strong accuracy but the DP variants remain informative.

Model and Algorithm. We follow the framework of Kairouz et al. (2021), adapting its training and accounting pipeline to our continual-release setting. We use a small MLP for MNIST/EMNIST and VGG-128 (Simonyan & Zisserman, 2015) for CIFAR-10. Our algorithms instantiate Algorithm 2 with different choices of the prefix-sum

²Our source code is available at:
<https://github.com/MengshiZ/ICML2026>.

mechanism PrivStreamSum, which determines how noisy gradients are aggregated over time:

- (i) `sgd_nodp`: non-private baseline without noise;
- (ii) `sgd_naive_dp`: i.i.d. Gaussian noise to each batch;
- (iii) `sgd_tree_dp`: standard tree-based DP prefix-sum;
- (iv) `sgd_matrix_dp`: DP prefix-sum based on the lower-triangular matrix mechanism of Fichtenberger et al. (2023).

We evaluate each method with and without the randomized buffering wrapper RandBin: without RandBin the stream is treated under Hamming-style neighbors, while with RandBin we obtain single-edit privacy. We append `_hamming` or `_edit` to indicate whether RandBin is disabled or enabled.

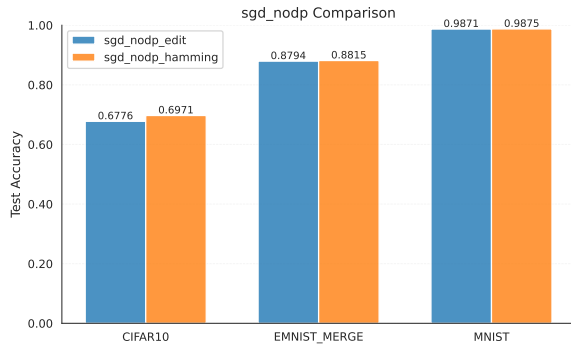


Figure 2. Comparison of `sgd_nodp_hamming` and `sgd_nodp_edit` baselines across MNIST, CIFAR-10, and EMNIST. This demonstrates that the RandBin wrapper incurs negligible accuracy overhead when converting from Hamming-style to single-edit privacy across all datasets.

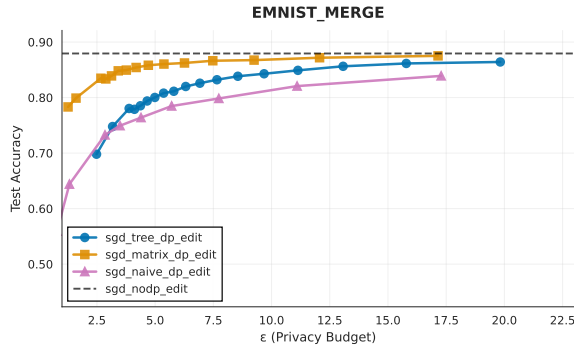


Figure 3. Accuracy-privacy tradeoff on EMNIST ByMerge with single-edit privacy. All algorithms use the RandBin wrapper (`sgd_naive_dp_edit`, `sgd_tree_dp_edit`, and `sgd_matrix_dp_edit`). Structured aggregation mechanisms improve over naive per-minibatch noise; MNIST and CIFAR-10 results are deferred to the appendix.

Privacy Parameter Setup. Instead of fixing (ϵ, δ) in advance, we treat the noise standard deviation as a hyperparameter and track RDP during training, then convert it to

(ϵ, δ) at a fixed δ ; see Section F.1 for the details. By sweeping the noise multiplier, we obtain accuracy-privacy trade-off curves. We fix $\delta = 10^{-5}$ for MNIST/CIFAR-10 and $\delta = 10^{-6}$ for EMNIST.

Implementation Details. We compare performance with and without RandBin. When RandBin is enabled, batch sizes vary in $[U, 2U]$; when it is disabled, we use a fixed batch size equal to the mean, $3U/2$, to match expected computation. For EMNIST and CIFAR-10 we set the mean batch size to 500, and for MNIST to 250.

Single-Edit Privacy Cost. Enabling RandBin (single-edit privacy) introduces only negligible accuracy loss relative to the corresponding Hamming-style baselines (`sgd_nodp_hamming` vs. `sgd_nodp_edit`), as shown in Figure 2 across all three datasets. This validates our core contribution: the randomized buffering approach achieves stronger privacy guarantees (single-edit versus Hamming neighbors) with nearly no utility cost.

Discussion of Results. Figure 3 presents the accuracy-privacy tradeoff on EMNIST ByMerge, for all three aggregation mechanisms under single-edit privacy. Similar patterns hold for MNIST and CIFAR-10, deferred to Appendix G.3. Across datasets we observe two consistent trends. First, as ϵ increases, the tree mechanism increasingly outperforms the naive per-minibatch noise approach, consistent with findings in Kairouz et al. (2021). Second, the matrix mechanism of Fichtenberger et al. (2023) exhibits the most robust performance across the entire privacy spectrum, providing the best accuracy-privacy tradeoffs.

7. Conclusion

We studied continual learning under adaptive interaction when privacy is defined over *edit-style* (single-insertion/deletion) user streams, where naive continual-release analyses do not directly apply. Our main recipe composes a randomized buffering wrapper RandBin, which reduces single-edit neighbors to a Hamming-style per-bin update interface with explicit backlog guarantees, with a certified continual-release DP primitive (tree-based prefix sums in our experiments), and then invokes modular composition to obtain end-to-end (ϵ, δ) -DP.

Empirically, the privacy target determines $U(\epsilon_b, \delta_b)$ and thus a concrete delay regime, and we observe the resulting utility-latency tradeoff across privacy levels. Beyond streaming SGD, the same buffering-aggregation interface can instantiate other optimizers and continual primitives (e.g., ADMM-style updates), which we defer to Appendix G.4. Further discussion on limitations is given in Appendix I.

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Impact Statement

Scope. This paper develops an auditable recipe for *continual-release* differential privacy in streaming/federated learning loops under *single-edit* participation changes. We focus on direct and tractable impacts arising from (i) protecting user participation over time and (ii) the explicit privacy–latency link induced by buffering.

Potential Benefits. Our mechanisms can reduce privacy risks in systems that repeatedly publish intermediate model updates (or derived statistics) by providing a principled trajectory-level guarantee under adaptive feedback. This may improve the safety of deploying learning systems in sensitive domains (e.g., health, finance, and user-facing personalization) and can support compliance efforts by making privacy guarantees more modular and checkable.

Potential Risks and Limitations. (1) *Misuse or overclaiming:* practitioners might treat the guarantee as a blanket protection against all forms of leakage (e.g., membership or attribute inference) even when deployment deviates from the assumed threat model, composition accounting, or noise calibration. (2) *Privacy washing:* stronger stated guarantees could be used to justify collecting more data or increasing retention without sufficient governance. (3) *Systems impact:* buffering introduces delay/backlog; in real deployments this may reduce responsiveness, create incentives to bypass privacy mechanisms, or shift risk to other system components. (4) *Residual risk:* differential privacy bounds expected leakage but does not eliminate it; careful parameter selection and monitoring remain necessary.

Uncertainties. Downstream impact depends on how the recipe is integrated into end-to-end pipelines (client sampling, clipping, aggregation, accounting) and on operational constraints that may differ across deployments.

Initiatives. We encourage (i) clear documentation of assumptions and privacy parameters, (ii) deployment checklists/tests that certify the required modular conditions (fresh per-round randomness, stable one-round privacy under common context, correct composition accounting, and threat-model alignment), and (iii) empirical evaluation of privacy–utility–latency trade-offs to discourage unsafe workarounds.

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Appendix

A. Detailed Preliminaries

We formalize the setting as outlined in the introduction. We first describe the concept of an *abstract* interactive mechanism, from which we will derive other forms of interactive mechanisms later.

Finite-Precision Convention. The simulation framework below is stated for finite message and seed spaces. For every public finite horizon T , we interpret all communicated inputs, states, and released numerical quantities using fixed finite-precision encodings. Whenever we write a continuous or unbounded-noise mechanism, the formal mechanism is obtained by applying a fixed, data-independent quantization and clipping map to the ideal noisy release. The resulting finite-output kernels admit finite-seed representations. By data processing, all stated DP, RDP, and zCDP upper bounds for the ideal releases continue to hold. We suppress this representation layer throughout.

Abstract Interactive Mechanism. We use $\mathcal{U}$ to denote the collection of all valid messages, where $\{\perp, \text{halt}\} \subseteq \mathcal{U}$. We use $\mathcal{R}$ to denote the collection of random seeds. In this work, we focus on the case that both $\mathcal{U}$ and $\mathcal{R}$ are finite.

An interactive mechanism $\mathcal{M}$ is specified by a distribution $R_{\mathcal{M}}$ on $\mathcal{R}$ and a transition function $\mathcal{M} : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$, where $\mathcal{U}^* := \cup_{i \in \mathbb{N}} \mathcal{U}^i$, with the convention that $\mathcal{U}^0$ is the identity under Cartesian product, i.e., $\mathcal{R} \times \mathcal{U}^0 = \mathcal{R}$. Note that we sometimes overload the notation such that $\mathcal{M}$ denotes both the mechanism and the corresponding transition function. Here is the functionality of an interactive mechanism.

1. The mechanism $\mathcal{M}$ samples a (secret) random seed $r \in \mathcal{R}$ according to the distribution $R_{\mathcal{M}}$.
2. At time step $t = 0$, the mechanism outputs $\mathcal{M}(r) \in \mathcal{U}$. If the mechanism is supposed to wait for the first input message, we use the convention that $\mathcal{M}(r) = \perp$.
3. At time step $t \geq 1$, suppose a message $x_t \in \mathcal{U}$ (which may be chosen depending on the history before time t) is sent to the mechanism. We use $x[1..t] := (x_1, x_2, \dots, x_t) \in \mathcal{U}^t$ to denote the messages received by the mechanism from the previous time steps up to t .
Then, at this time step, the mechanism outputs $\mathcal{M}(r; x[1..t]) \in \mathcal{U}$.
4. If, for some t , $\mathcal{M}(r; x[1..t]) = \text{halt}$, then the mechanism halts at time step t .

Instantiation: Mapping the Abstract Interaction to Our Transcript. The abstract interaction formalism above subsumes the continual-learning setting of Section 2. Concretely, we instantiate the mechanism’s input messages as the user-side stream $x_t := f_t$ (or, when modeling a subcomponent, x_t may represent an emitted bin B_t or a per-bin update g_t), and we instantiate the mechanism’s outputs as the released observations $y_t := (v_t, w^{(t)})$, so that the induced transcript is $\text{tr}_{\leq T} = ((v_t, w^{(t)}))_{t=0}^T$. The adversary’s view function $\nu_{\mathcal{A}}$ reveals exactly the observable portion of each release—the timing bit and model snapshot—matching the trajectory-and-timing threat model in Section 2.

Bounded Termination. In addition to finite $\mathcal{R}$ and $\mathcal{U}$, we consider mechanisms that always terminate in a bounded number of steps. Specifically, for each mechanism $\mathcal{M}$, there exists $T_{\mathcal{M}} \in \mathbb{Z}$ such that $\mathcal{M}$ always terminates in at most $T_{\mathcal{M}}$ steps. This is without loss of generality, because any privacy notion defined for mechanisms with bounded termination can be extended naturally to unbounded mechanisms. We simply require that, for any $T \in \mathbb{Z}_+$, the truncated mechanism obtained by running the (unbounded) mechanism for T steps will satisfy the privacy notion defined for bounded mechanisms. This matches the “for every horizon T ” convention in Section 2.

Mechanism View. The view observed by a mechanism $\mathcal{M}$ up to time step t consists of its generated random seed r and its received input messages $x[1..t]$ up to time t .

Remark A.1. Since $\mathcal{M}$ can recover its own output sequence from its random seed and input sequence, it suffices to include the latter two in its view.

Similarity Between Mechanisms. To formalize privacy notions later, we need a way to quantify the similarity between two mechanisms. Since we consider randomized mechanisms, we will use power functions to compare how different

two distributions are. It is known that power functions are general enough to capture all divergences satisfying the data processing inequality. Recall that in this work, we focus on finite sample spaces.

Definition A.2 (Data Processing Inequality). A divergence measure is a function D that takes two distributions P and Q on the same space such that $D(P\|Q) \geq 0$, where equality holds *iff* the distributions $P = Q$ are identical.

A divergence D satisfies the *data processing inequality* if, for any stochastic transformation (or channel) T that maps the original space to another space, the following inequality holds:

$$D(T(P)\|T(Q)) \leq D(P\|Q),$$

where $T(P)$ and $T(Q)$ are the resulting distributions after applying the transformation T to P and Q , respectively.

Definition A.3 (Power Function as Fractional Knapsack Problem (Kadane, 1968)). Suppose P and Q are distributions on the same finite sample space Ω , i.e., P and Q are vectors in $\mathbb{R}_{\geq 0}^{\Omega}$ whose coordinates sum to 1. The power function $\text{Pow}(P\|Q) : [0, 1] \rightarrow [0, 1]$ can be defined in terms of the *fractional knapsack problem*.

Given a collection Ω of items, suppose $\omega \in \Omega$ has weight $P(\omega)$ and value $Q(\omega)$. Then, given $\alpha \in [0, 1]$, $\text{Pow}(P\|Q)(\alpha)$ is the maximum value attained with weight capacity constraint α , where items may be taken fractionally.

Intuition. When two distributions P and Q are the same, it is clear that given any capacity constraint α , the maximum reward is also α ; this means the power function is exactly the identity function. However, if the two distributions are very different, this means that there are items whose reward-to-weight ratios are large; hence, in this case, the power function can initially grow faster than the identity function.

Partial Order on Power Functions. Pointwise comparison naturally induces a partial order on power functions. We denote $f_1 \leq f_2$ if for all $\alpha \in [0, 1]$, $f_1(\alpha) \leq f_2(\alpha)$, where a larger f_2 indicates that the two distributions are more different.

Fact A.4 (Properties of Power Functions). For any two distributions P and Q on the same sample space, $\text{Pow}(P\|Q) \geq \text{Id}$, where $\text{Id} : [0, 1] \rightarrow [0, 1]$ is the identity function, and equality holds *iff* the distributions $P = Q$ are identical.

Moreover, power functions satisfy the data processing inequality, i.e., for any stochastic transformation (or channel) T that maps the original space to another space, the following inequality holds:

$$\text{Pow}(T(P)\|T(Q)) \leq \text{Pow}(P\|Q).$$

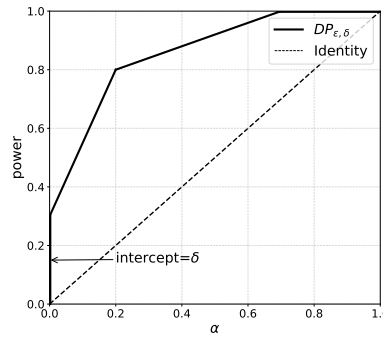


Figure 4. Power function $\text{DP}_{\epsilon, \delta}$ for $\epsilon = 1$ and $\delta = 0.3$.

Differential Privacy in Terms of Power Function. Given $\epsilon \geq 0$ and $0 \leq \delta \leq 1$, the power function $\text{DP}_{\epsilon, \delta} : [0, 1] \rightarrow [0, 1]$ can be described as follows (see Figure 4):

1. In the xy -plane, starting at $(0, \delta)$, the function increases linearly with slope e^ϵ until it touches the line $y = 1 - x$ at $(\frac{1-\delta}{1+e^\epsilon}, \frac{e^\epsilon + \delta}{1+e^\epsilon})$.
2. After touching the line $y = 1 - x$, the slope changes to $e^{-\epsilon}$, until the line segment reaches $(1 - \delta, 1)$, where the curve remains horizontally till the end $(1, 1)$ is reached.

In other words,

$$\text{DP}_{\varepsilon, \delta}(\alpha) = \begin{cases} \delta + e^\varepsilon \cdot \alpha, & \text{for } 0 \leq \alpha \leq \frac{1-\delta}{1+e^\varepsilon}; \\ 1 - e^{-\varepsilon}(1-\delta) + e^{-\varepsilon}\alpha, & \text{for } \frac{1-\delta}{1+e^\varepsilon} < \alpha \leq 1-\delta; \\ 1, & \text{for } 1-\delta < \alpha \leq 1. \end{cases}$$

Below are some facts on power functions that we will use, which are usually equivalently stated in the literature in terms of the tradeoff function $\mathsf{T}(P\|Q)(\alpha) := 1 - \text{Pow}(P\|Q)(\alpha)$.

Fact A.5 (DP in Terms of Power Function (Dong et al., 2022)). *Given two distributions P and Q on the same sample space Ω , $\text{Pow}(P\|Q) \leq \text{DP}_{\varepsilon, \delta}$ is equivalent to the statement that for all $S \subseteq \Omega$, $P(S) \leq e^\varepsilon \cdot Q(S) + \delta$ and $Q(S) \leq e^\varepsilon \cdot P(S) + \delta$.*

Fact A.6. (Triangle Inequality for Power Functions)[Theorem 2.14 in (Dong et al., 2022)] *Suppose X, Y, Z are distributions on the same sample space such that $\text{Pow}(X\|Y) \leq g_1$ and $\text{Pow}(Y\|Z) \leq g_2$. Then, $\text{Pow}(X\|Z) \leq g_2 \circ g_1$, where the composition is defined as $(g_2 \circ g_1)(x) := g_2(g_1(x))$. In other words, $\text{Pow}(X\|Z) \leq \text{Pow}(Y\|Z) \circ \text{Pow}(X\|Y)$.*

Example. Consider $g = \text{DP}_{\varepsilon, \delta}$. Then, $g \circ g \leq \text{DP}_{2\varepsilon, \delta(1+e^\varepsilon)}$. As we shall see, this is relevant to the scenario of *multi-hop* neighbors – aka *group DP* in the literature.

Definition A.7 (Supremum of Power Functions). Given a collection $\mathcal{S}$ of power functions, its supremum $\sup(\mathcal{S}) = \sup_{f \in \mathcal{S}} f$ is the least power function $\hat{f}$ such that for all $f \in \mathcal{S}$, $f \leq \hat{f}$.

Formally, for each $\alpha \in [0, 1]$, $\hat{f}(\alpha) :=$

$$\sup\{\lambda \cdot f_1(\alpha_1) + (1-\lambda) \cdot f_2(\alpha_2) \mid f_1, f_2 \in \mathcal{S}; \alpha_1, \alpha_2, \lambda \in [0, 1] : \alpha = \lambda\alpha_1 + (1-\lambda)\alpha_2\}.$$

Definition A.8 (Tensor Product of Power Functions). Suppose f and g are power functions that are attained by the corresponding distributions in $\text{Pow}(P_1\|P_2) = f$ and $g = \text{Pow}(Q_1\|Q_2)$. Then, their corresponding tensor product $f \otimes g = \text{Pow}(P_1 \times Q_1\|P_2 \times Q_2)$ is defined in terms of the corresponding product distributions.

Fact A.9 (Joint Convexity (Dong et al., 2022)). *Suppose Λ is an indexing set and f is a power function such that for each $\lambda \in \Lambda$, a pair $(P(\lambda), Q(\lambda))$ of distributions satisfy $\text{Pow}(P(\lambda)\|Q(\lambda)) \leq f$.*

Then, for any distribution L on Λ , $\text{Pow}(P(L)\|Q(L)) \leq f$.

Fact A.10 (Composition Rule of Power Functions (Dong et al., 2022)). *Suppose P and P' are distributions on the sample space Ω . For each $\omega \in \Omega$, $Q(\omega)$ and $Q'(\omega)$ are distributions on sample space Ω' . If $\text{Pow}(P\|P') \leq f$ and $\text{Pow}(Q(\omega)\|Q'(\omega)) \leq g$ for all $\omega \in \Omega$, then $\text{Pow}((P, Q(P))\|(P', Q'(P')))) \leq f \otimes g$.*

Distinguishing between Interactive Mechanisms. Suppose we wish to compare the behavior of two mechanisms $\mathcal{M}_0$ and $\mathcal{M}_1$, to which we only have oracle accesses. An adversary $\mathcal{A} : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$ is also an interactive mechanism. The adversary has some view function $\nu_{\mathcal{A}} : \mathcal{U} \rightarrow \mathcal{U}$ that specifies what information it can gather from the output of the mechanism. However, later when we instantiate from the abstract mechanism, we will use the structure of the message space to clarify what the adversary can observe.

We next define the interaction $\mathcal{A} \leftrightarrow \mathcal{M}$ between the two mechanisms. The adversary $\mathcal{A}$ repeatedly sends messages to $\mathcal{M}$. At each time step, the adversary determines the next message to be sent to $\mathcal{M}$, based on the history of prior interactions. This process repeats iteratively, until the mechanism $\mathcal{M}$ outputs halt in some time step. Formally, it is captured by the following description.

1. The mechanisms $\mathcal{A}$ and $\mathcal{M}$ sample the random seeds $r_{\mathcal{A}}$ and $r_{\mathcal{M}}$, respectively.
 2. At time step $t = 0$, $\mathcal{M}$ outputs $y_0 \leftarrow \mathcal{M}(r_{\mathcal{M}}) \in \mathcal{U}$.
 $\mathcal{A}$ observes $v_0 := \nu_{\mathcal{A}}(y_0)$.
 3. At time step $t \geq 1$,
 - (a) $\mathcal{A}$ generates $x_t \leftarrow \mathcal{A}(r_{\mathcal{A}}; v[0..t-1])$ and sends to $\mathcal{M}$.
 - (b) $\mathcal{M}$ responds $y_t \leftarrow \mathcal{M}(r_{\mathcal{M}}; x[1..t])$;
 $\mathcal{A}$ observes $v_t := \nu_{\mathcal{A}}(y_t)$.
- The process terminates as soon as either x_t or y_t is halt; otherwise, the process continues for the next time step.

Definition A.11 (Adversary View). Suppose the interaction $\mathcal{A} \leftrightarrow \mathcal{M}$ terminates at time step t (which may be a random quantity). Then, the adversary view is defined as:

$$\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}) := (r_{\mathcal{A}}; v[0..t]).$$

Terminology (“non-adaptive”). In this paper, when we say “non-adaptive” for a continual mechanism’s privacy analysis, we mean the mechanism’s input stream is fixed in advance (equivalently, the environment/adversary that provides inputs does not condition on past outputs).

Definition A.12 (Non-Adaptive Adversary). An adversary transition function $\mathcal{A} : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$ is non-adaptive, if there exists a transition function of the form $\hat{\mathcal{A}} : \mathcal{R} \times \mathbb{Z} \rightarrow \mathcal{U}$ such that for any length- t sequence $v[0..t-1]$, $\mathcal{A}(r; v[0..t-1]) = \hat{\mathcal{A}}(r; t)$.

We can next define a distance notion between two mechanisms based on power functions. Intuitively, it quantifies the difficulty of distinguishing between two mechanisms.

Definition A.13 (Power Function between Mechanisms). Given two interactive mechanisms $\mathcal{M}_0$ and $\mathcal{M}_1$, we overload the notation and denote $\text{Pow}(\mathcal{M}_0 \parallel \mathcal{M}_1) \leq f$, for some power function f , if for all adversaries $\mathcal{A}$, $\text{Pow}(\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_0) \parallel \text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_1)) \leq f$.

If we have the weaker requirement that the inequality holds for just non-adaptive adversaries, we denote: $\text{Pow}^{\text{NA}}(\mathcal{M}_0 \parallel \mathcal{M}_1) \leq f$.

Simulation of Two Close Interactive Mechanisms with Close Random Seeds. The following result from (Vadhan & Zhang, 2023) states that if two mechanisms are close in the sense of Definition A.13, then it is possible to simulate either one of them exactly with the same mechanism transition function that initially takes a random seed generated from one of the two appropriate distributions. We use the finite-space, bounded-termination version of this result. Under the finite-precision convention above, every bounded-horizon mechanism considered in this paper falls within its scope.

This is a very useful fact in proving privacy composition results, because instead of interacting with two potentially different mechanisms in every time step, we can view the adversary as interacting with the same mechanism transition function but with initial random seeds drawn from two different distributions. One useful analogy is that after a sensitive database is anonymized with appropriate noise, then it can be queried unlimited number of times without privacy degradation because of data post-processing.

Fact A.14 (Exact Simulation of Two Close Interactive Mechanisms (Vadhan & Zhang, 2023), Theorem 1.5). Suppose $\mathcal{M}_0$ and $\mathcal{M}_1 : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$ are two interactive mechanisms with bounded termination, and f is a power function such that $\text{Pow}(\mathcal{M}_0 \parallel \mathcal{M}_1) \leq f$ in the sense of Definition A.13.

Then, there exist two distributions N_0 and N_1 on some sample space Ω and a transition function $\mathcal{P} : \Omega \times \mathcal{U}^* \rightarrow \mathcal{U}$ such that $\text{Pow}(N_0 \parallel N_1) \leq f$, and, for all adversaries $\mathcal{A}$, the following holds for both $b \in \{0, 1\}$: $\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_b) \equiv \text{view}(\mathcal{A} \leftrightarrow \mathcal{P}(N_b))$,

where $\mathcal{P}(N)$ is the interactive mechanism that uses the specified transition function $\mathcal{P}$ with a random seed drawn from the distribution N .

B. Modular Composition of Interactive Mechanisms with Hidden Outputs

Interactive Mechanism with Hidden Output. We next consider a special form of interactive mechanism with a transition function of the form $\mathcal{M} : \mathcal{R} \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{Y}$. Here, we use $\mathcal{X}$ to denote the input space. In each time step, the mechanism produces $(v, y) \in \mathcal{V} \times \mathcal{Y}$, where the adversary can observe v from the view space $\mathcal{V}$ and the output y from the output space $\mathcal{Y}$ is potentially hidden from the adversary, which may be passed to another interactive mechanism (that could however possibly leak information about y). If an adversary only interacts with $\mathcal{M}$, then it can be described by a transition function with the form $\mathcal{A} : \mathcal{R} \times \mathcal{V}^* \rightarrow \mathcal{X}$.

Neighboring Input Sequences. Similar to the conventional differential privacy, we need a neighboring notion for input sequences such that intuitively the behaviors of a mechanism on two neighboring input sequences should be similar.

Assumption B.1 (Prefix-Closed Neighboring Relation). We assume there is a neighboring relation $\sim_{\mathcal{X}^*}$ (or just $\sim$ for succinctness) that satisfies the property that if two sequences $x[1..t] \sim x'[1..t']$ are neighboring, then the following holds:

- (i) They have the same length, i.e., $t = t'$.
- (ii) All their corresponding prefixes are neighboring, i.e., for all $1 \leq \tau < t$, $x[1..\tau] \sim x'[1..\tau]$.

Privacy for Interactive Mechanisms. Since the inputs for an interactive mechanism in later time steps can be influenced by its earlier behavior, one cannot formally define its privacy just based on neighboring input sequences. In the literature, *adaptive* differential privacy (Jain et al., 2023) is based on a construction that we name as *paired simulation*.

Definition B.2 (Paired Simulation). Suppose an interactive mechanism is given by the transition function $\mathcal{M} : \mathcal{R} \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{Y}$ and the random seed distribution $R_{\mathcal{M}}$, where a neighboring relation is defined on $\mathcal{X}^*$.

Then, the paired simulation consists of a canonical pair of interactive mechanisms $(\mathcal{M}_0^{\text{pair}}, \mathcal{M}_1^{\text{pair}})$ such that for each $b \in \{0, 1\}$, the transition function has the form $\mathcal{M}_b^{\text{pair}} : \mathcal{R} \times (\mathcal{X}^2)^* \rightarrow \mathcal{V} \times \mathcal{Y}$. Moreover, in the paired simulation, an adversary with transition function $\mathcal{A} : \mathcal{R} \times \mathcal{V}^* \rightarrow \mathcal{X}^2$ interacts with $\mathcal{M}_b^{\text{pair}}$ as follows.

The interaction between an adversary $\mathcal{A}$ and $\mathcal{M}_b^{\text{pair}}$:

1. $\mathcal{A}$ and $\mathcal{M}$ sample the hidden random seeds $r_{\mathcal{A}}$ and $r_{\mathcal{M}}$ from their own distributions, respectively.
2. At time step $t = 0$, the simulation calls $(v_0, y_0) \leftarrow \mathcal{M}(r_{\mathcal{M}})$. If $\mathcal{M}$ is supposed to wait for the first input message, $(v_0, y_0) = (\perp, \perp)$.
3. At time step $t \geq 1$, $\mathcal{A}$ constructs a pair $(x_t^{(0)}, x_t^{(1)}) \leftarrow \mathcal{A}(r_{\mathcal{A}}; v[0..t-1]) \in \mathcal{X}^2$ and sends to the simulation.
If the sequences $x^{(0)}[1..t]$ and $x^{(1)}[1..t]$ generated so far are not neighboring, the whole simulation terminates.
Otherwise, $(v_t, y_t) \leftarrow \mathcal{M}(r_{\mathcal{M}}; x^{(b)}[1..t])$, and $\mathcal{A}$ observes v_t .
If either $\mathcal{A}$ or $\mathcal{M}$ decides to halt, then the whole simulation terminates.

The adversarial view is denoted as $\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_b^{\text{pair}})$. If the simulation does not terminate before time step t , then the view up to time t includes the random seed $r_{\mathcal{A}}$ and $v[0..t]$.

Definition B.3 (Differentially Private (DP) View). Given an interactive mechanism $\mathcal{M}$, suppose $(\mathcal{M}_0^{\text{pair}}, \mathcal{M}_1^{\text{pair}})$ is the paired simulation as in Definition B.2. For a power function f , the mechanism $\mathcal{M}$ is adaptively f -differentially private (DP) if

$\text{Pow}(\mathcal{M}_0^{\text{pair}} \parallel \mathcal{M}_1^{\text{pair}}) \leq f$, for all adaptive adversaries in the sense of Definition A.13. Similarly, non-adaptively f -DP is defined by $\text{Pow}^{\text{NA}}(\mathcal{M}_0^{\text{pair}} \parallel \mathcal{M}_1^{\text{pair}}) \leq f$.

Multi-Hop Neighboring Notion. In the literature, group DP or multi-hop neighbors can be generalized to sequences readily.

Definition B.4 (k -Hop Neighboring Sequences). For $k = 1$, 1-hop neighboring is the same as that in Assumption B.1. For $k \geq 2$, two sequences x and x' are k -hop neighboring if there exist $x_1, \dots, x_{k-1}$ such that for $0 \leq i < k$, x_i and x_{i+1} are 1-hop neighboring, where $x_0 = x$ and $x_k = x'$.

We also generalize Definition B.2 to multi-hop neighboring simulation.

Definition B.5 (*k*-Hop Neighboring Simulation). Generalizing the description in Definition B.2, in each time step t , the adversary generates a tuple $(x_t^{(0)}, x_t^{(1)}, \dots, x_t^{(k)})$ such that for each $0 \leq i < k$, $x^{(i)}[1..t]$ and $x^{(i+1)}[1..t]$ are neighboring sequences.

Moreover, for $b \in [0..k]$, $\mathcal{M}_b^{\text{hop}}$ is a simulation of $\mathcal{M}$ that takes the b -th component $x_t^{(b)}$ at each step t .

Fact B.6 (Adaptive DP for Multi-Hop Neighboring Sequences). Suppose $\mathcal{M}$ is adaptively f -DP as in Definition B.3. Then, in a k -hop neighboring simulation as in Definition B.5, we have: $\text{Pow}(\mathcal{M}_0^{\text{hop}} \| \mathcal{M}_k^{\text{hop}}) \leq f^{\circ k}$,

where $f^{\circ 1} = f$ and $f^{\circ(i+1)} = f^{\circ i} \circ f$.

Proof. Fix a k -hop adaptive adversary, and let P_i denote the distribution of its view in simulation $\mathcal{M}_i^{\text{hop}}$. For every $i < k$, projecting each adversarially generated tuple onto coordinates $(i, i+1)$ gives a valid adaptive paired-simulation adversary. Hence adaptive one-hop f -DP gives

$$\text{Pow}(P_i \| P_{i+1}) \leq f.$$

Applying the triangle inequality for power functions repeatedly yields

$$\text{Pow}(P_0 \| P_k) \leq f^{\circ k}.$$

□

Modular Composition of Interactive Mechanisms. As mentioned above, the hidden output of one mechanism may be fed as the input of another mechanism. To this end, we define a specific form of composition that we call *modular composition*. The adversary only directly supplies an input to the “head” mechanism of the list, where each intermediate mechanism receives its input from the (hidden) output from the previous mechanism.

Modular composition is an interactive mechanism with a transition function of the form $(\mathcal{M} \rightarrow \mathcal{N}) : \mathcal{R}_1 \times \mathcal{R}_2 \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{W} \times \mathcal{Z}$ described as follows.

1. The mechanisms $\mathcal{A}$, $\mathcal{M}$ and $\mathcal{N}$ sample the hidden random seeds $r_{\mathcal{A}}$, $r_{\mathcal{M}}$, $r_{\mathcal{N}}$ from their own distributions, respectively.
2. At time step $t = 0$, call the subroutine $(v_0, y_0) \leftarrow \mathcal{M}(r_{\mathcal{M}})$ and $(w_0, z_0) \leftarrow \mathcal{N}(r_{\mathcal{N}})$.
The adversary observes (v_0, w_0) .

3. At time step $t \geq 1$,

- (a) the adversary generates $x_t \leftarrow \mathcal{A}(r_{\mathcal{A}}; v[0..t-1], w[0..t-1])$, and sends it to the head mechanism of the modular composition $(\mathcal{M} \rightarrow \mathcal{N})$.
- (b) call the subroutine $(v_t, y_t) \leftarrow \mathcal{M}(r_{\mathcal{M}}; x[1..t])$ and pass the hidden output y_t to the next mechanism $\mathcal{N}$;
- (c) call the subroutine $(w_t, z_t) \leftarrow \mathcal{N}(r_{\mathcal{N}}; y[1..t])$;
- (d) the adversary observes (v_t, w_t) .

The whole process terminates as soon as one of $\mathcal{A}$, $\mathcal{M}$ and $\mathcal{N}$ terminates. Otherwise, the process continues for the next time step.

If the process terminates at time step t , the view of the adversary consists of its random seed $r_{\mathcal{A}}$, $v[0..t]$ and $w[0..t]$.

Definition B.7 (Modular Composition). Suppose two interactive mechanisms are given by the transition functions $\mathcal{M} : \mathcal{R}_1 \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{Y}$ and $\mathcal{N} : \mathcal{R}_2 \times \mathcal{Y}^* \rightarrow \mathcal{W} \times \mathcal{Z}$, with random seed distributions $R_{\mathcal{M}}$ and $R_{\mathcal{N}}$, respectively. Then, an adversary with transition function $\mathcal{A} : \mathcal{R} \times (\mathcal{V} \times \mathcal{W})^* \rightarrow \mathcal{X}$ interacts with the modular composition $\mathcal{M} \rightarrow \mathcal{N}$ as above.

Neighbor-Preserving Paired Simulation. To consider composition of differentially oblivious mechanisms (Chan et al., 2022), the framework neighbor-preserving differential obliviousness (NPDO) (Zhou et al., 2023; 2024) has been introduced to formalize the condition that if neighboring inputs are submitted to a mechanism, then the corresponding outputs would somehow be neighboring for a subsequent mechanism. To generalize this to interactive mechanisms, we will need to

introduce neighbor-preserving paired simulation. In addition to neighboring input sequences in $\mathcal{X}^*$ for mechanism $\mathcal{M}$, we assume that there is a similar notion of neighboring sequences in $\mathcal{Y}^*$ (satisfying Assumption B.1) for mechanism $\mathcal{N}$.

Definition B.8 (Neighbor-Preserving Paired Simulation (NPP)). Given an interactive mechanism with transition function $\mathcal{M} : \mathcal{R} \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{Y}$ and random seed distribution $R_{\mathcal{M}}$, a neighbor-preserving paired simulation (which might not be unique) is a pair $(\mathcal{M}_0^{\text{npp}}, \mathcal{M}_1^{\text{npp}})$ of interactive mechanisms such that the following holds.

1. There exists possibly another collection $\mathcal{R}'$ of random seeds and an appropriate distribution $R'_{\mathcal{M}}$ such that for each $b \in \{0, 1\}$, the corresponding mechanism has a transition function of the form $\mathcal{M}_b^{\text{npp}} : \mathcal{R} \times \mathcal{R}' \times (\mathcal{X}^2)^* \rightarrow \mathcal{V} \times \mathcal{Y}^2$, where the random seed distribution on $\mathcal{R}$ still follows $R_{\mathcal{M}}$, and the random seed from $\mathcal{R}'$ is sampled from $R'_{\mathcal{M}}$.

In other words, in addition to a random seed in $\mathcal{R}$, an extra random seed in $\mathcal{R}'$ may be taken. Just like paired simulation in Definition B.2, a sequence of input pairs are also taken. However, the transition function returns a tuple in $\mathcal{V} \times \mathcal{Y}^2$, i.e., in addition to $v \in \mathcal{V}$, a pair $(y_0, y_1) \in \mathcal{Y}^2$ is returned.

2. For each $b \in \{0, 1\}$, $r \in \mathcal{R}$, $r' \in \mathcal{R}'$, $x^{(0)}[1..t]$ and $x^{(1)}[1..t] \in \mathcal{X}^*$, if $x^{(0)}[1..t]$ and $x^{(1)}[1..t]$ are not neighboring sequences, then $\mathcal{M}_b^{\text{npp}}(r, r'; x^{(0)}[1..t], x^{(1)}[1..t]) = \perp$, and the whole simulation terminates at time step t .

Otherwise, the pair $(y^{(0)}[1..t], y^{(1)}[1..t])$ of output sequences produced in the first t time steps are neighboring in $\mathcal{Y}^*$.

3. If for some $(v, y_0, y_1) \in \mathcal{V} \times \mathcal{Y}^2$, $\mathcal{M}_b^{\text{npp}}(r, r'; x^{(0)}[1..t], x^{(1)}[1..t]) = (v, y_0, y_1)$, then $\mathcal{M}(r; x^{(b)}[1..t]) = (v, y_b)$.

In other words, we require consistency between $\mathcal{M}_b^{\text{npp}}$ and $\mathcal{M}_b^{\text{pair}}$ (from Definition B.2). Specifically, the projection of $\mathcal{M}_b^{\text{npp}}$'s output onto $\mathcal{M}_b^{\text{pair}}$'s range must match the behavior of $\mathcal{M}_b^{\text{pair}}$.

However, unlike paired simulation $\mathcal{M}_b^{\text{pair}}$ whose behavior is totally determined by the given $\mathcal{M}$, in the tuple (v, y_0, y_1) returned above by $\mathcal{M}_b^{\text{npp}}$, the value $y_{\bar{b}}$ (where $b \neq \bar{b}$) can depend on all the input parameters in a manner that is not specified by $\mathcal{M}$.

4. **Augmented Adversary View.** When an adversary interacts with $\mathcal{M}_b^{\text{npp}}$, the adversarial view consists of its randomness and the sequence in $(\mathcal{V} \times \mathcal{Y}^2)^*$ produced until termination.

Definition B.9 (NPDP Interactive Mechanisms). Given a power function f , an interactive mechanism $\mathcal{M}$ is said to be adaptively f -neighbor-preserving differentially private (NPDP), if there exists a neighbor-preserving paired simulation $(\mathcal{M}_0^{\text{npp}}, \mathcal{M}_1^{\text{npp}})$ such that $\text{Pow}(\mathcal{M}_0^{\text{npp}} \| \mathcal{M}_1^{\text{npp}}) \leq f$, in the sense of Definition A.13.

Remark B.10 (Construction of Neighbor-Preserving Paired Simulation). Observe that if one wishes to prove that a mechanism is NPDP as in Definition B.9, there is potentially some freedom to construct an appropriate NPP simulation.

On the contrary, for non-interactive mechanisms (that only run for one time step), the recent result on universally closest distribution refinements (Chan & Xue, 2025) implies that a “best possible” NPP simulation may be used.

Unfortunately, this does not seem to generalize to interactive mechanisms. The high level reason is that there may be no closest distribution refinements on two random sequences of length $t + 1$ such that its prefix projection would give a closest distribution refinement for the corresponding prefixes of length t . In other words, if one constructs the “best possible” NPP up to time step t , it may not result in the best NPP if the process enters step $t + 1$.

C. Modular Composition Theorem for Adaptively Differentially Private Interactive Mechanisms

We next describe a modular composition theorem for interactive mechanisms. At a high level, the theorem states that if the first stage is *neighbor-preserving* (NPDP) with privacy profile f , and the second stage is (adaptive) DP with profile g on the intermediate stream, then the end-to-end interactive pipeline is (adaptive) DP with profile $f \otimes g$.

How this theorem relates to existing composition frameworks. Our theorem can be viewed as a direct consequence of two existing ideas, once NPDP is interpreted through *paired simulation*.

First, it is shown (Zhou et al., 2024) that neighbor-preserving notions (including differential obliviousness/NPDO) admit an equivalent characterization in terms of a paired simulation that outputs a *refinement pair*. Informally, in the standard

secret-bit distinguishing game, an adversary provides at each time step a neighboring pair $(x_t^{(0)}, x_t^{(1)})$, and a secret bit $b \in \{0, 1\}$ selects which component is the “real” input. A mechanism is DP if the adversary’s views under $b = 0$ and $b = 1$ are close. The refinement-pair view resolves the usual obstacle in modular composition: even if the real world uses $x^{(b)}$ and produces a real intermediate output $y^{(b)}$, one also needs a synthetic neighbor $\bar{y}^{(1-b)}$ so that the next stage can be analyzed in a paired simulation. The NPDP guarantee precisely asserts the existence of such a paired simulator that, for each b , produces a pair $(y^{(0)}, y^{(1)})$ whose b -marginal matches the real execution, and such that the two resulting pair-distributions (for $b = 0$ vs. $b = 1$) are close.

Second, a general framework for adaptive privacy of continual mechanisms (Henzinger et al., 2026) is developed to prove concurrent composition theorems under interactive adversaries. In particular, once a mechanism is expressed in the standard secret-bit game (and the controller only interacts with downstream submechanisms through their privatized outputs), privacy of the overall system follows from standard interactive composition and post-processing. Under the paired-simulation perspective, our two-stage pipeline $\mathcal{M} \rightarrow \mathcal{N}$ fits this template: the NPDP stage can be viewed as a mechanism that, on secret bit b , outputs a paired intermediate stream $(y_t^{(0)}, y_t^{(1)})$ (together with visible outputs v_t), and the DP stage $\mathcal{N}$ is then run on $y^{(b)}$. Thus, one may reduce modular composition to ordinary interactive composition: compose the lifted NPDP mechanism (as DP-on-the-bit with profile f) with the second-stage DP mechanism (profile g), and finally apply post-processing to remove the extra synthetic transcript.

While this reduction gives a useful conceptual guide and could be formalized by instantiating the conditions of (Henzinger et al., 2026) together with the refinement-pair characterization of (Zhou et al., 2024), we include below a self-contained proof tailored to our setting and our power-function accounting. In particular, a more general concurrent setting (multiple submechanisms with interleaving calls) is studied in (Henzinger et al., 2026), whereas we only need the serial modular composition used by our buffering–aggregation pipeline. Providing a direct proof also makes explicit how the privacy profile tensor $f \otimes g$ arises in our notation, and it cleanly interfaces with the certification arguments in other sections.

Theorem C.1. *Suppose an interactive mechanism with transition function $\mathcal{M} : \mathcal{R}_1 \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{Y}$ is adaptively f -NPDP (as in Definition B.9), and another one with $\mathcal{N} : \mathcal{R}_2 \times \mathcal{Y}^* \rightarrow \mathcal{W} \times \mathcal{Z}$ is adaptively g -DP (as in Definition B.3). Then, the modular composition (as in Definition B.7) $\mathcal{M} \rightarrow \mathcal{N} : \mathcal{R}_1 \times \mathcal{R}_2 \times \mathcal{X}^* \rightarrow \mathcal{V} \times \mathcal{W} \times \mathcal{Z}$ is adaptively $f \otimes g$ -DP.*

The augmented adversary $\hat{\mathcal{A}}$ interacts with $\hat{\mathcal{L}}_b$, for each $b \in \{0, 1\}$, as follows.

1. The mechanisms $\mathcal{A}$, $(\mathcal{M}_0^{\text{npp}}, \mathcal{M}_1^{\text{npp}})$ and $(\mathcal{N}_0^{\text{pair}}, \mathcal{N}_1^{\text{pair}})$, sample the random seeds $r_{\mathcal{A}}$, $(r_{\mathcal{M}}, r'_{\mathcal{M}})$, $r_{\mathcal{N}}$ from the appropriate distributions, respectively.
2. At time step $t = 0$, call the subroutines $(v_0, y_0^{(0)}, y_0^{(1)}) \leftarrow \mathcal{M}_b^{\text{npp}}(r_{\mathcal{M}}, r'_{\mathcal{M}})$ and $(w_0, z_0) \leftarrow \mathcal{N}_b^{\text{pair}}(r_{\mathcal{N}})$.
 $\hat{\mathcal{A}}$ observes (v_0, w_0) .
3. At time step $t \geq 1$,
 - (a) $\hat{\mathcal{A}}$ generates $(x_t^{(0)}, x_t^{(1)}) \leftarrow \mathcal{A}(r_{\mathcal{A}}; v[0..t-1], w[0..t-1])$, and sends it to $\hat{\mathcal{L}}_b$.
 - (b) $(v_t, y_t^{(0)}, y_t^{(1)}) \leftarrow \mathcal{M}_b^{\text{npp}}(r_{\mathcal{M}}, r'_{\mathcal{M}}; x^{(0)}[1..t], x^{(1)}[1..t])$ and pass the hidden output pair $(y_t^{(0)}, y_t^{(1)})$ to the next mechanism $\mathcal{N}_b^{\text{pair}}$;
 - (c) $(w_t, z_t) \leftarrow \mathcal{N}_b^{\text{pair}}(r_{\mathcal{N}}; y^{(0)}[1..t], y^{(1)}[1..t])$;
 - (d) $\hat{\mathcal{A}}$ observes (v_t, w_t) .

The whole process terminates as soon as one of $\mathcal{A}$, $\mathcal{M}_b^{\text{npp}}$ and $\mathcal{N}_b^{\text{pair}}$ terminates. Otherwise, the process continues for the next time step.

If the process terminates at time step t , the view of the adversary consists of its random seed $r_{\mathcal{A}}$, $v[0..t]$ and $w[0..t]$.

Proof. We denote the resulting interactive mechanism from the modular composition by $\mathcal{L} := \mathcal{M} \rightarrow \mathcal{N}$. In view of Definition B.3, we consider the paired simulation $(\mathcal{L}_0^{\text{pair}}, \mathcal{L}_1^{\text{pair}})$ as described in Definition B.2. Specifically, our goal is to show that for any adversary $\mathcal{A}$, the two distributions $\text{view}(\mathcal{A} \leftrightarrow \mathcal{L}_0^{\text{pair}})$ and $\text{view}(\mathcal{A} \leftrightarrow \mathcal{L}_1^{\text{pair}})$ are “close” in the sense quantified by the power function $f \otimes g$.

Our proof strategy is to construct an augmented pair $(\hat{\mathcal{L}}_0, \hat{\mathcal{L}}_1)$ of interactive mechanisms and an augmented adversary $\hat{\mathcal{A}}$ whose views have the same distributions as those from $(\mathcal{L}_0^{\text{pair}}, \mathcal{L}_1^{\text{pair}})$ and $\mathcal{A}$.

We next use Fact A.14 to show that instead of considering the pair $(\mathcal{L}_0^{\text{pair}}, \mathcal{L}_1^{\text{pair}})$ with the same random seed distribution, we can equivalently simulate the pair with some common transition function, but using two different random seed distributions. By the data processing inequality, it suffices to analyze the alternative simulation and the power function of the two random seed distributions. Because $\mathcal{M}$ is f -NPDP, let $(\mathcal{M}_0^{\text{npp}}, \mathcal{M}_1^{\text{npp}})$ be the corresponding NPP simulation as guaranteed by Definition B.9 such that $\text{Pow}(\mathcal{M}_0^{\text{npp}} \| \mathcal{M}_1^{\text{npp}}) \leq f$. By Fact A.14, there exist some interactive mechanism $\mathcal{P}$ and two random seed distributions F_0 and F_1 such that $\text{Pow}(F_0 \| F_1) \leq f$ and for each $b \in \{0, 1\}$, $\mathcal{P}(F_b)$ is an exact simulation of $\mathcal{M}_b^{\text{npp}}$ (with the appropriate random seed distribution $R_{\mathcal{M}} \times R'_{\mathcal{M}}$) for any adversary. Specifically, for each b , $\mathcal{M}_b^{\text{npp}}(R_{\mathcal{M}} \times R'_{\mathcal{M}})$ and $\mathcal{P}(F_b)$ are equivalent.

Similarly, since $\mathcal{N}$ is g -DP, the paired simulation $(\mathcal{N}_0^{\text{pair}}, \mathcal{N}_1^{\text{pair}})$ satisfies $\text{Pow}(\mathcal{N}_0^{\text{pair}} \| \mathcal{N}_1^{\text{pair}}) \leq g$. Again, by Fact A.14, there exist some interactive mechanism $\mathcal{Q}$ and two random seed distributions G_0 and G_1 such that $\text{Pow}(G_0 \| G_1) \leq g$ and for each $b \in \{0, 1\}$, $\mathcal{Q}(G_b)$ is an exact simulation of $\mathcal{N}_b^{\text{pair}}$ (with the appropriate random seed distribution $R_{\mathcal{N}}$) for any adversary. Specifically, for each b , $\mathcal{N}_b^{\text{pair}}(R_{\mathcal{N}})$ and $\mathcal{Q}(G_b)$ are equivalent.

Given an adversary $\mathcal{A}$ that interacts with $\mathcal{L}_b^{\text{pair}}$ for $b \in \{0, 1\}$, we describe how an augmented adversary $\hat{\mathcal{A}}$ interacts with $\hat{\mathcal{L}}_b$ in the above colored box. Because we follow exactly the same structure as in Definition B.7, after comparing the two descriptions line-by-line, it follows that $\text{view}(\mathcal{A} \leftrightarrow \mathcal{L}_b^{\text{pair}})$ has the same distribution as $\text{view}(\hat{\mathcal{A}} \leftrightarrow \hat{\mathcal{L}}_b)$.

As aforementioned, we use Fact A.14, for each $b \in \{0, 1\}$, we can equivalently replace $\mathcal{M}_b^{\text{npp}}(R_{\mathcal{M}}, R'_{\mathcal{M}})$ with $\mathcal{P}(F_b)$, and $\mathcal{N}_b^{\text{pair}}(R_{\mathcal{N}})$ with $\mathcal{Q}(G_b)$ in the above description.

Hence, the first inequality below follows from the data processing inequality:

$$\begin{aligned} \text{Pow}(\text{view}(\mathcal{A} \leftrightarrow \mathcal{L}_0^{\text{pair}}) \| \text{view}(\mathcal{A} \leftrightarrow \mathcal{L}_1^{\text{pair}})) &= \text{Pow}(\text{view}(\hat{\mathcal{A}} \leftrightarrow \hat{\mathcal{L}}_0) \| \text{view}(\hat{\mathcal{A}} \leftrightarrow \hat{\mathcal{L}}_1)) \\ &\leq \text{Pow}(F_0 \times G_0 \| F_1 \times G_1) \\ &\leq f \otimes g, \end{aligned}$$

where the second inequality follows from the data processing inequality and the exact simulations, and the last inequality follows from Fact A.10. $\square$

D. RandBin: Achieving Privacy for Edit-Style Neighboring Inputs

Continual-release mechanisms are usually analyzed under Hamming-style neighboring streams. Given a collection $\mathcal{O}$ of objects, an infinite input stream is a map $x : \mathbb{Z}_{\geq 1} \rightarrow \mathcal{O}$. Two streams $x \sim_H x'$ are Hamming-neighboring if they differ in at most one index. In our setting, however, participation itself is sensitive, so one stream may instead be obtained from the other by inserting or deleting one item.

Definition D.1 (Neighboring in the Edit Sense). Given a stream x and $t \in \mathbb{Z}_{\geq 1}$, let $\text{Delete}(x; t)$ be the stream x' defined by

$$x'[\tau] = \begin{cases} x[\tau], & \tau < t, \\ x[\tau + 1], & \tau \geq t. \end{cases}$$

Two streams $x \sim_E x'$ are neighboring in the edit sense if one is obtained from the other by one such deletion. Two finite sequences of the same length are edit-neighboring if they are prefixes of two edit-neighboring infinite streams.

D.1. Motivating Example: Private Prefix Sums Under Edit Adjacency

For an integer stream σ , the prefix-sum workload releases estimates of $\sum_{i=1}^t \sigma[i]$ for every t . Standard tree-based mechanisms are designed for Hamming neighbors, where one bounded coordinate changes between the two streams (Dwork et al., 2010a; Chan et al., 2011). An edit can instead shift every subsequent coordinate. For example, deleting the first element of

$$(1, 0, 1, 0, \dots)$$

produces $(0, 1, 0, 1, \dots)$, so the two streams differ at every position. Thus the usual one-coordinate sensitivity analysis does not apply directly.

The role of RandBin is to bridge these neighboring notions. It buffers arriving items and occasionally emits a random-sized prefix of the buffer as one concrete bin. Consider the neighboring streams

$$\sigma = [a, b, c, d, e, f], \quad \sigma' = [b, c, d, e, f, g].$$

One possible pair of binnings is

$$\begin{aligned} \sigma &: [\perp, \perp, \{a, b\}, \perp, \perp, \{c, d, e\}], \\ \sigma' &: [\perp, \perp, \{b\}, \perp, \perp, \{c, d, e\}]. \end{aligned}$$

The emitted sequences now differ in only one concrete bin. The formal construction below makes the release schedules identical and confines the effect of one edit to at most two adjacent concrete bins, while ensuring that the two paired executions have close distributions.

D.2. Technical Description of RandBin

We first define the structured Hamming-style relation used at the output interface.

Definition D.2 (Neighboring Bin Sequences). For the parameter U fixed below, let $\mathcal{B}$ be the collection of non-empty concrete bins, each containing an array of at most $2U + 1$ objects. A bin sequence of length T is a map

$$B : [1..T] \rightarrow \mathcal{B} \cup \{\perp\},$$

where $\perp$ denotes that no concrete bin is emitted. Two concrete bins at times $i < j$ are adjacent if all outputs strictly between them are $\perp$.

Two bin sequences $B \sim_{\mathcal{B}} B'$ are neighboring if they have the same concrete-bin emission times and differ in at most two positions. More precisely:

1. If they differ in exactly one position, that position is the last concrete bin in both sequences, and the two bins have the form

$$[\sigma_1, w, \sigma_2] \quad \text{and} \quad [\sigma_1, \sigma_2, u].$$

2. If they differ in exactly two positions, these positions contain adjacent concrete bins in both sequences, and the two pairs have the form

$$\begin{aligned} &[\sigma_1, w, \sigma_2], \perp, \dots, \perp, [u, \sigma_3], \\ &[\sigma_1, \sigma_2, u], \perp, \dots, \perp, [\sigma_3], \end{aligned}$$

for items w, u and possibly empty arrays $\sigma_1, \sigma_2, \sigma_3$.

Identical bin sequences are also neighboring. In every case, concatenating the concrete bins gives finite prefixes of edit-neighboring item streams.

Thus $\sim_{\mathcal{B}}$ is a structured distance-at-most-two relation with respect to ordinary per-bin Hamming adjacency. If a downstream mechanism is analyzed only for one-coordinate Hamming neighbors, its distance-two, or group-DP, profile must be used when invoking the modular composition theorem.

Definition D.3 (Symmetric Geometric Distribution). For $\alpha > 1$, the symmetric geometric distribution $\text{Geom}(\alpha)$ on $\mathbb{Z}$ has probability mass function

$$\Pr \text{Geom}(\alpha) = k = \frac{\alpha - 1}{\alpha + 1} \alpha^{-|k|}.$$

Definition D.4 (Truncated Geometric Distribution). Let $\varepsilon > 0$ and $\delta \in (0, 1)$. Let U be the smallest even positive integer such that

$$\Pr |\text{Geom}(e^\varepsilon)| \geq U/2 \leq \delta.$$

In particular,

$$U = O\left(\frac{1}{\varepsilon} \log \frac{1}{\delta}\right).$$

The truncated geometric distribution $G(\varepsilon, \delta)$ is obtained by sampling $R \sim \text{Geom}(e^\varepsilon)$ and returning

$$\text{clip}(R, [-U/2, U/2]) := \min\{\max\{R, -U/2\}, U/2\}.$$

Algorithm 3 uses the prefix-sum mechanism $\text{StreamSum}_{\varepsilon_2, \delta_2}$ from Theorem F.2, instantiated with $[a..b] = [U..2U]$. Write

$$E[i] := E_{\varepsilon_2, \delta_2}(i).$$

The theorem gives all properties needed here: the mechanism is adaptively $(\varepsilon_2, \delta_2)$ -DP for unit-Hamming integer streams; on every actual load stream in $[U..2U]$, it satisfies

$$\left| S[i] - \sum_{j=1}^i C[j] \right| \leq E[i], \quad S[i] - S[i-1] \in [U..2U]$$

simultaneously for every i and every realization of its randomness.

Algorithm 3 Interactive RandBin Mechanism

- 1: **Input:** $\varepsilon > 0, \delta \in (0, 1)$; a stream $x : \mathbb{Z}_{\geq 1} \rightarrow \mathcal{O}$
- 2: **Output:** A bin sequence $\mathbb{Z}_{\geq 1} \rightarrow \mathcal{B} \cup \{\perp\}$; the visible output reveals only whether a concrete bin or $\perp$ is emitted
- 3: Set $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$ and $\delta_1 = \delta_2 = \delta/2$
- 4: Let U be the value fixed by Definition D.4
- 5: Let

$$\mathcal{G} := \frac{3U}{2} + G(\varepsilon_1, \delta_1),$$

which has support $[U..2U]$

- 6: Independently initialize $\text{StreamSum}_{\varepsilon_2, \delta_2}$ from Theorem F.2 for the interval $[U..2U]$, and set $E[i] := E_{\varepsilon_2, \delta_2}(i)$
 - 7: Initialize an empty FIFO queue $\text{Buf} \leftarrow \emptyset$
 - 8: Set $\text{Index} \leftarrow 0, C[0] \leftarrow 0, S[0] \leftarrow 0, E[0] \leftarrow 0$, and $\text{Bin}[0] \leftarrow \perp$
 - 9: **for** each step $t \geq 1$ at which $x[t]$ is received **do**
 - 10: Append $x[t]$ to Buf
 - 11: **if** $t < S[\text{Index}] + E[\text{Index}] + 2U$ **then**
 - 12: Output $\perp$
 - 13: **else**
 - 14: If $C[\text{Index}] = 0$, set $\text{Bin}[\text{Index}] \leftarrow \perp$; otherwise, remove the first $C[\text{Index}]$ items from Buf to form $\text{Bin}[\text{Index}]$
 - 15: Output $\text{Bin}[\text{Index}]$
 - 16: $\text{Index} \leftarrow \text{Index} + 1$
 - 17: Sample $C[\text{Index}] \sim \mathcal{G}$ independently
 - 18: $S[\text{Index}] \leftarrow \text{StreamSum}_{\varepsilon_2, \delta_2}(C[\text{Index}])$
 - 19: **end if**
 - 20: **end for**
-

Theorem D.5 (Utility Guarantee of RandBin). *At the end of every step t , for every realization of the randomness, the concatenation of the concrete bins emitted so far is a prefix of the input stream, and the number of items remaining in Buf is*

$$O\left(1 + \frac{1}{\varepsilon} \log(t+1) \left(\log(t+1) + \log \frac{1}{\delta}\right)\right).$$

Equivalently, the emitted prefix has length at least t minus this quantity.

Proof. Let

$$A_i := \sum_{j=1}^i C[j], \quad A_0 := 0.$$

Suppose bin $i \geq 1$ is emitted at step t . The emission condition and the always-valid error bound imply

$$t \geq S[i] + E[i] + 2U \geq A_i + 2U.$$

Immediately before removing bin i , the queue therefore contains

$$t - A_{i-1} \geq C[i] + 2U$$

items, so underflow is impossible. Since the queue is FIFO, the emitted items always form a prefix of the input stream.

At the end of every step, the current index Index satisfies

$$t < S[\text{Index}] + E[\text{Index}] + 2U. \quad (1)$$

This is immediate if no bin is emitted. If a bin is emitted, the previous threshold is crossed for the first time at the current integer step, while

$$S[\text{Index} + 1] - S[\text{Index}] \geq U \geq 1 \quad \text{and} \quad E[\text{Index} + 1] \geq E[\text{Index}],$$

so the next threshold is strictly larger than t .

If $\text{Index} \geq 1$, the number of buffered items at the end of step t is $t - A_{\text{Index}-1}$. Using (1), consistency, and the error bound,

$$\begin{aligned} t - A_{\text{Index}-1} &< S[\text{Index}] + E[\text{Index}] + 2U - (S[\text{Index} - 1] - E[\text{Index} - 1]) \\ &\leq 4U + E[\text{Index}] + E[\text{Index} - 1]. \end{aligned}$$

For $\text{Index} = 0$, it is simply $t < 2U$. Since $\text{Index} \leq t$, Definition D.4 and Theorem F.2 give the claimed bound. $\square$

D.3. Adaptive Privacy Analysis

The visible release schedule of RandBin depends only on its sampled loads and the prefix-sum scheduler, not on the item values. Nevertheless, the concrete bins are hidden outputs passed to the downstream mechanism. We must therefore control the joint distribution of the visible schedule and these hidden outputs, as required by adaptive NPDP in Definition B.9.

The NPP simulation. Recall from Definition B.8 that an NPP simulator receives paired edit-neighboring input prefixes $(x^{(0)}, x^{(1)})$. In branch b , it must simulate the real execution on $x^{(b)}$ and simultaneously construct a synthetic output sequence for $x^{(1-b)}$ so that the two output sequences are $\sim_B$ -neighboring. The construction below makes explicit the paired refinement underlying the static RandBin analysis in (Zhou et al., 2023).

As a standard bookkeeping convention, we assume that each logical item appears at most once and carries a unique tag, preserved across the two neighboring streams and ignored by downstream computation. Thus, if the two input streams first differ at step τ , then at the beginning of step $\tau + 1$ the simulator can determine which stream contains the unmatched item w . This one-step delay is harmless: the $2U$ slack ensures that w cannot be emitted at step τ .

Definition D.6 (NPP Simulation for RandBin). For $b \in \{0, 1\}$, $\text{RandBin}_b^{\text{npp}}$ runs Algorithm 3 on $x^{(b)}$, producing the visible release schedule and the real bin sequence $y^{(b)}$. It uses the same release times to construct the synthetic sequence $y^{(1-b)}$.

Until the unmatched item affects an emitted bin, the two bin sequences are identical. Suppose first that the real branch is the longer input stream, and let the real adjacent concrete bins containing the unmatched item w and the following item u be

$$[\sigma_1, w, \sigma_2], \perp, \dots, \perp, [u, \sigma_3].$$

The simulator outputs in the synthetic sequence

$$[\sigma_1, \sigma_2, u], \perp, \dots, \perp, [\sigma_3].$$

If the second concrete bin has not yet been emitted, only the first replacement is made, and the second is deferred. If the real branch is the shorter stream, the simulator applies the inverse transformation. The construction for $b = 1$ is symmetric.

Consequently, the two output sequences have identical release times and are neighboring according to Definition D.2.

The construction is online. Indeed, suppose the first affected bin is bin J . At its emission time t ,

$$t \geq S[J] + E[J] + 2U \geq A_J + 2U.$$

Thus, after removing its $C[J]$ items, at least one further item u is already available in the buffer.

We now isolate the two privacy components: the shifted fresh load and the private prefix-sum scheduler.

Lemma D.7 (Adaptive Privacy of RandBin). *Let $Z \sim \mathcal{G}$. Suppose that, for some power function f_{load} ,*

$$\text{Pow}(Z \| Z - \theta) \leq f_{\text{load}} \quad \text{for both } \theta \in \{-1, +1\}. \quad (2)$$

Suppose also that the prefix-sum scheduler is adaptively f_{sum} -DP for unit-Hamming integer streams and uses randomness independent of the load samples. Then the NPP simulation in Definition D.6 satisfies

$$\text{Pow}(\text{RandBin}_0^{\text{npp}} \| \text{RandBin}_1^{\text{npp}}) \leq f_{\text{load}} \otimes f_{\text{sum}}.$$

Hence RandBin is adaptively $(f_{\text{load}} \otimes f_{\text{sum}})$ -NPDP.

Proof. Fix a finite horizon T . By joint convexity (Fact A.9), it suffices to fix a deterministic adaptive adversary $\mathcal{A}$. We analyze the stronger view that also reveals the paired bin lengths and every prefix-sum output $S[k]$ when it is generated. The original NPP transcript is a post-processing of this view.

If no edit occurs, the two branches are identical. Otherwise, let τ be the first step at which the input prefixes differ, and let i be the current bin index immediately before the item at step τ is appended. If $i = 0$, this item cannot enter bin 0 because $C[0] = 0$. If $i \geq 1$, bin $i - 1$ was emitted at some step $r < \tau$. At the beginning of step τ , before appending the new item, the buffer therefore contains at least

$$r - A_{i-1} \geq S[i - 1] + E[i - 1] + 2U - A_{i-1} \geq 2U$$

older items. Since $C[i] \leq 2U$, the new item cannot enter bin i .

Let J be the first affected concrete-bin index. The preceding argument gives $J \geq i + 1$. The simulator preserves the length of bin J by replacing w with u ; hence the paired length streams $D^{(0)}, D^{(1)}$ differ only at

$$K := J + 1,$$

where their values differ by exactly 1. The deletion direction is known at the beginning of step $\tau + 1$, whereas at most one bin can be emitted in any step. Thus bin J cannot be emitted before the direction is known. Moreover, Algorithm 3 samples $C[K]$ only after bin J has been emitted. Consequently, both K and the sign

$$\theta := D^{(1)}[K] - D^{(0)}[K] \in \{-1, +1\}$$

are determined before $C[K]$ is sampled.

Let $\mathcal{F}_k$ be the complete augmented history immediately before $C[k]$ is sampled. Since the loads are fresh and independent,

$$\text{Law}(C[k] \mid \mathcal{F}_k) = \mathcal{G}. \quad (3)$$

For $k \neq K$, the paired lengths satisfy

$$(D^{(0)}[k], D^{(1)}[k]) = (C[k], C[k])$$

in both branches. At coordinate K , they satisfy

$$(D^{(0)}[K], D^{(1)}[K]) = \begin{cases} (C[K], C[K] + \theta), & \text{in branch 0,} \\ (C[K] - \theta, C[K]), & \text{in branch 1.} \end{cases}$$

Thus, conditioned on every reachable pre- K history, the first coordinate has law Z in branch 0 and law $Z - \theta$ in branch 1. In either branch, the second coordinate is obtained from the first by adding θ . Hence the paired release is obtained from the corresponding scalar experiment through the same injective map

$$a \mapsto (a, a + \theta),$$

whose inverse on its image is projection onto the first coordinate. Its conditional power function is therefore exactly

$$\text{Pow}(Z \| Z - \theta) \leq f_{\text{load}}$$

by (2).

All earlier and later load kernels are common to the two branches. By the composition rule for power functions, joint convexity, and data processing, the interactive paired-length generator is adaptively f_{load} -NPDP. This conclusion remains valid although K is selected from previous scheduler outputs, because the conditional bound holds at every reachable pre- K history.

The streams $D^{(0)}$ and $D^{(1)}$ are unit-Hamming neighbors. Their synthetic coordinates may lie in $[U - 1..2U + 1]$, but Theorem F.2 defines the scheduler and guarantees adaptive privacy on all integer streams; the interval $[U..2U]$ is needed only for the execution-wise consistency guarantee on the actual load stream. Applying the modular composition theorem (Theorem C.1) to the paired-length generator and the adaptive prefix-sum scheduler gives the joint profile

$$f_{\text{load}} \otimes f_{\text{sum}}.$$

Finally, the input pairs, paired lengths, and scheduler outputs determine the complete NPP transcript by the FIFO rule: the scheduler outputs determine the emission times, and the paired lengths determine the two bin contents. As shown above, the $2U$ slack makes this reconstruction causal. The claimed bound follows by data processing. Since T and $\mathcal{A}$ were arbitrary, the result holds for the unbounded interaction. $\square$

Corollary D.8. *The interactive mechanism RandBin in Algorithm 3 is adaptively (ε, δ) -NPDP with input neighboring relation $\sim_E$ and output neighboring relation $\sim_B$.*

Proof. Let $Z \sim \mathcal{G}$ and fix $\theta \in \{-1, +1\}$. Away from the atoms inherited from the two clipped tails, the likelihood ratio between Z and $Z - \theta$ is at most e^{ε_1} . The exceptional atoms under Z have total mass at most

$$\Pr |\text{Geom}(e^{\varepsilon_1})| \geq U/2 \leq \delta_1.$$

Hence

$$\text{Pow}(Z \| Z - \theta) \leq \text{DP}_{\varepsilon_1, \delta_1} \quad \text{for both } \theta \in \{-1, +1\}.$$

By Theorem F.2, the scheduler is adaptively $(\varepsilon_2, \delta_2)$ -DP for unit-Hamming integer streams. Lemma D.7 and basic composition now give

$$\begin{aligned} \text{Pow}(\text{RandBin}_0^{\text{NPP}} \| \text{RandBin}_1^{\text{NPP}}) &\leq \text{DP}_{\varepsilon_1, \delta_1} \otimes \text{DP}_{\varepsilon_2, \delta_2} \\ &\leq \text{DP}_{\varepsilon_1 + \varepsilon_2, \delta_1 + \delta_2} = \text{DP}_{\varepsilon, \delta}. \end{aligned}$$

$\square$

E. Adaptive Safety via Quantitative Decomposability

This section proves the adaptive-safety certification used in the modular pipeline. The goal is to identify structural conditions under which an ordinary non-adaptive privacy analysis of a continual mechanism already implies privacy against adaptive adversaries. The result applies to Hamming-style neighboring input streams and is stated in the power-function language used throughout the appendix.

The first condition is a fresh-randomness, prefix-causality condition. At round t , the mechanism may depend on the entire input prefix $x[1..t]$, but the only randomness used for the round- t output is a fresh seed ω_t independent of all other rounds. This rules out hidden cross-round correlations introduced by reused noise, while still allowing the release at time t to be an arbitrary deterministic function of the input prefix.

Definition E.1 (Independently Decomposable Mechanisms). An (abstract) interactive mechanism with transition function $\mathcal{M} : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$ is independently decomposable if the following properties hold.

- Each random seed $r \in \mathcal{R}$ takes the form $r = (\omega_1, \omega_2, \dots) \in \Omega^{\mathbb{N}}$. Moreover, the mechanism samples ω_t independently from some distribution R_t in time step t .
- For each $t \geq 1$, there exists a deterministic function $\mathcal{M}_t$ such that the output $y_t \leftarrow \mathcal{M}_t(\omega_t; x[1..t]) \in \mathcal{U}$ depends only on $x[1..t]$ and the randomness ω_t sampled in step t .

Quantitatively Decomposable Mechanisms. Independent decomposability alone records where the randomness enters, but does not quantify how much a change in the input prefix can affect a one-round release. We therefore add a quantitative condition. The condition says that, for each round t , the power function between the two possible round- t output laws is determined exactly by the distance between two aggregates of selected input coordinates. The invariant-pair condition below ensures that common aggregation context can be ignored: only the part of the aggregate containing the Hamming discrepancy contributes to the distance.

Definition E.2 (Invariant Pair). Let $d : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}_{\geq 0}$ be a metric, and let $(\mathcal{U}, \oplus, 0_\oplus)$ be a monoid. We say that $(d, \oplus)$ is an invariant pair (under common aggregation context) if for all $a, b, u, u' \in \mathcal{U}$,

$$d(a \oplus u \oplus b, a \oplus u' \oplus b) = d(u, u').$$

Notation. For an index set $I = \{\tau_1 < \tau_2 < \dots < \tau_k\} \subseteq [1..t]$ and an input history $x[1..t] \in \mathcal{U}^t$, define

$$\bigoplus_{\tau \in I} x[\tau] := x[\tau_1] \oplus x[\tau_2] \oplus \dots \oplus x[\tau_k].$$

For $I = \emptyset$, we use the convention $\bigoplus_{\tau \in I} x[\tau] := 0_\oplus$. Since $\oplus$ is associative and the elements are aggregated in increasing index order, this expression is well-defined without requiring commutativity.

Definition E.3 (Quantitatively Decomposable Mechanism). An interactive mechanism $\mathcal{M} : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$ is *quantitatively decomposable* with respect to an invariant pair $(d, \oplus)$ in the sense of Definition E.2 if it is independently decomposable in the sense of Definition E.1, and there exist a sequence of index sets $\{I_t\}_{t \geq 1}$ with $I_t \subseteq [1..t]$ and, for every $t \geq 1$ and $\theta \geq 0$, a power function $g_t(\theta) : [0, 1] \rightarrow [0, 1]$ such that the following hold.

- For every $t \geq 1$, $g_t(\theta) \leq g_t(\theta')$ whenever $0 \leq \theta \leq \theta'$.
- For every $t \geq 1$ and every pair of input histories $x[1..t], x'[1..t] \in \mathcal{U}^t$,

$$\text{Pow}(\mathcal{M}_t(R_t; x[1..t]) \| \mathcal{M}_t(R_t; x'[1..t])) = g_t \left(d \left(\bigoplus_{\tau \in I_t} x[\tau], \bigoplus_{\tau \in I_t} x'[\tau] \right) \right).$$

Here, $\mathcal{M}_t(R_t; x[1..t])$ denotes the distribution of $\mathcal{M}_t(\omega_t; x[1..t])$ when ω_t is sampled from R_t .

We can now state the certification theorem. The only-if direction is immediate, since non-adaptive adversaries are a special case of adaptive adversaries. The content is the converse: for quantitatively decomposable mechanisms, adaptivity does not increase the worst-case power-function profile. Thus any finite-horizon non-adaptive f -DP proof automatically lifts to the paired-simulation adaptive game.

Theorem E.4 (Adaptive Safety of Quantitatively Decomposable Mechanisms). *Consider the Hamming-style neighboring relation on equal-length input sequences in $\mathcal{U}$, where two sequences are neighboring if they differ in at most one position. Let f be a power function. Suppose $\mathcal{M} : \mathcal{R} \times \mathcal{U}^* \rightarrow \mathcal{U}$ is quantitatively decomposable in the sense of Definition E.3. If $\mathcal{M}$ is non-adaptively f -DP in the paired-simulation sense of Definition B.3, for every finite horizon, then $\mathcal{M}$ is adaptively f -DP in the same sense, for every finite horizon.*

Proof. We prove the following finite-horizon statement by induction on T : for every power function f , if $\mathcal{M}$ is non-adaptively f -DP up to horizon T , then $\mathcal{M}$ is adaptively f -DP up to horizon T . This implies the theorem because the theorem assumes the non-adaptive guarantee for every finite horizon.

We first reduce to deterministic adversaries. A randomized adversary can be viewed as first sampling its random seed and then running a deterministic adversary. For every fixed seed, the deterministic-adversary case gives the desired power-function bound. Taking the mixture over the same seed distribution under the two secret bits preserves the bound by the joint convexity property in Fact A.9. Hence it suffices to consider deterministic adaptive adversaries.

We also assume, without loss of generality, that the adversary observes the entire output of $\mathcal{M}$ in each round. If the real adversary observes only a view of the output, the real view is obtained from the full-output transcript by post-processing, and the desired bound follows from data processing. If the mechanism has an initial output before the adversary sends any input, this initial output has the same distribution under the two secret bits; it may be included as an additional common first observation and does not affect the privacy bound. Thus we count rounds from the first adversarial input.

Let $\mathcal{A}$ be a deterministic adaptive adversary in the paired simulation $(\mathcal{M}_0^{\text{pair}}, \mathcal{M}_1^{\text{pair}})$. We pad terminated executions with a fixed halt symbol, so all transcripts have length T . This padding is post-processing and does not change the privacy claim. We must prove

$$\text{Pow}\left(\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_0^{\text{pair}}) \parallel \text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_1^{\text{pair}})\right) \leq f.$$

The base case $T = 1$ is immediate. With only one input round, the adversary has no previous mechanism output on which to adapt, so the adaptive and non-adaptive paired simulations coincide.

Assume the statement holds for horizon $T - 1$, and consider horizon $T > 1$. Let

$$(x^{(0)}[1], x^{(1)}[1])$$

be the first pair of inputs produced by $\mathcal{A}$. Since $\mathcal{A}$ is deterministic, this pair is fixed. Let

$$U^{(b)} := \mathcal{M}_1(R_1; x^{(b)}[1])$$

be the first-round output distribution under secret bit b .

We split into two cases.

Case 1: the first round is the challenge round. Assume $x^{(0)}[1] \neq x^{(1)}[1]$, and write

$$\Delta := d(x^{(0)}[1], x^{(1)}[1]).$$

Because the paired simulation only allows Hamming-neighboring prefixes, once the first inputs differ, every later input pair produced by the adversary must have equal components. That is, on every non-terminated transcript prefix and for every $t \geq 2$, the adversary must output a pair of the form (x_t, x_t) . We use the finite-horizon convention that terminated executions are padded with a fixed halt symbol; the padded rounds have identical outputs under the two secret bits and hence contribute the identity power function.

For each round t , define

$$h_t := \begin{cases} g_t(\Delta), & \text{if } 1 \in I_t, \\ \text{Id}, & \text{if } 1 \notin I_t. \end{cases}$$

We first show that, at every round t and every transcript prefix that can arise before round t , the conditional next-output distributions under the two secret bits have power function exactly h_t .

Fix such a transcript prefix. Since the adversary is deterministic, conditioning on this common prefix fixes all input pairs generated so far. As argued above, the two input histories entering round t differ only in the first coordinate. If $1 \notin I_t$, then the two aggregates

$$\bigoplus_{\tau \in I_t} x^{(0)}[\tau] \quad \text{and} \quad \bigoplus_{\tau \in I_t} x^{(1)}[\tau]$$

are identical. Hence their distance is zero. Applying Definition E.3 to identical histories gives $g_t(0) = \text{Id}$, and therefore the one-round power function is exactly Id .

If $1 \in I_t$, then the two aggregates have the form

$$x^{(0)}[1] \oplus c \quad \text{and} \quad x^{(1)}[1] \oplus c$$

for the same common suffix aggregate c over the remaining indices in I_t . By the invariant-pair property,

$$d(x^{(0)}[1] \oplus c, x^{(1)}[1] \oplus c) = d(x^{(0)}[1], x^{(1)}[1]) = \Delta.$$

Hence Definition E.3 gives the exact one-round identity

$$\text{Pow}\left(\mathcal{M}_t(R_t; x^{(0)}[1..t]) \parallel \mathcal{M}_t(R_t; x^{(1)}[1..t])\right) = g_t(\Delta) = h_t.$$

Thus the one-round power function is exactly h_t , and h_t depends only on t , not on the transcript prefix.

We now justify that these exact one-round identities tensorize to an exact identity for the whole adaptive transcript in this case. Let P_b be the distribution of the full length- T transcript under secret bit b . For a transcript prefix $s = y[1..t-1]$, let $K_{t,s}^b$ be the conditional distribution of y_t given the prefix s under bit b . By independent decomposability, $K_{t,s}^b$ is generated using the fresh seed R_t , which is independent of the previous seeds. From the preceding paragraph,

$$\text{Pow}(K_{t,s}^0 \| K_{t,s}^1) = h_t$$

for every prefix s that can arise.

We use the following finite-experiment characterization of power functions. For two finite distributions P and Q , the power function $\text{Pow}(P \| Q)$ is determined by the decreasing rearrangement of the likelihood ratio $Q(\omega)/P(\omega)$ under $\omega \sim P$; equivalently, two pairs of finite distributions with the same power function have the same likelihood ratio distribution under their first distribution, up to the usual splitting of atoms. Moreover, tensor product of power functions corresponds to taking the product of independent likelihood ratios.

Apply this characterization to the conditional kernels. For each prefix s , the likelihood ratio

$$L_{t,s}(y_t) := \frac{K_{t,s}^1(y_t)}{K_{t,s}^0(y_t)}$$

has, under $y_t \sim K_{t,s}^0$, the likelihood-ratio distribution determined by h_t . This distribution is the same for every prefix s . Since R_t is fresh and independent of the previous seeds, the conditional likelihood-ratio increment at round t is independent of the past after conditioning on the prefix, and its distribution is determined only by h_t .

For a full transcript $y[1..T]$, the likelihood ratio between P_1 and P_0 factors as

$$\frac{P_1(y[1..T])}{P_0(y[1..T])} = \prod_{t=1}^T \frac{K_{t,y[1..t-1]}^1(y_t)}{K_{t,y[1..t-1]}^0(y_t)}.$$

Under P_0 , the factors in this product have the same joint distribution as independent likelihood-ratio variables whose one-step power functions are $h_1, \dots, h_T$. Therefore the power function of the full adaptive transcript pair is exactly the tensor product:

$$\text{Pow}(P_0 \| P_1) = \bigotimes_{t=1}^T h_t = \bigotimes_{\substack{t \in [1..T] \\ 1 \in I_t}} g_t(\Delta),$$

where the second equality uses $\text{Id} \otimes q = q$ for every power function q .

Finally, consider any non-adaptive adversary that submits the fixed first pair $x^{(0)}[1], x^{(1)}[1]$ and then submits identical inputs in every later round. The later common inputs may be chosen arbitrarily. By the invariant pair property, they only contribute common aggregation context and therefore do not change the one-step power labels h_t . For this non-adaptive input pair, the same tensor product is the power function of the transcript pair:

$$\bigotimes_{\substack{t \in [1..T] \\ 1 \in I_t}} g_t(\Delta).$$

Since $\mathcal{M}$ is non-adaptively f -DP up to horizon T , this tensor product is at most f . Hence

$$\text{Pow}(\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_0^{\text{pair}}) \| \text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_1^{\text{pair}})) \leq f$$

in Case 1.

Case 2: the first round is not the challenge round. Assume $x^{(0)}[1] = x^{(1)}[1]$. Denote this common value by c . Then $U^{(0)} = U^{(1)}$ as distributions, and hence

$$\text{Pow}(U^{(0)} \| U^{(1)}) = \text{Id}.$$

We now define a shifted mechanism for the remaining $T-1$ rounds after the common first input c has been fixed. Let $\mathcal{M}^{(c)}$ be the mechanism whose round- s output, for $1 \leq s \leq T-1$, is

$$\mathcal{M}_s^{(c)}(R_{s+1}; z[1..s]) := \mathcal{M}_{s+1}(R_{s+1}; c, z[1..s]).$$

This mechanism uses the independent seeds $R_2, \dots, R_T$, so it is independently decomposable. It is also quantitatively decomposable. Indeed, for round s define

$$I_s^{(c)} := \{\tau - 1 : \tau \in I_{s+1}, \tau \geq 2\} \subseteq [1..s].$$

If $1 \notin I_{s+1}$, then the aggregate used by $\mathcal{M}_{s+1}$ is exactly the aggregate over $I_s^{(c)}$ in the suffix history. If $1 \in I_{s+1}$, then the aggregate has the common left context c , and the invariant-pair property removes this common context. Therefore, for every two suffix histories $z[1..s]$ and $z'[1..s]$,

$$\text{Pow}\left(\mathcal{M}_s^{(c)}(R_{s+1}; z[1..s]) \parallel \mathcal{M}_s^{(c)}(R_{s+1}; z'[1..s])\right) = g_{s+1} \left(d \left(\bigoplus_{\rho \in I_s^{(c)}} z[\rho], \bigoplus_{\rho \in I_s^{(c)}} z'[\rho] \right) \right).$$

Thus $\mathcal{M}^{(c)}$ satisfies the same structural assumptions as $\mathcal{M}$, with the time indices shifted by one.

Let Λ be the set of all Hamming-neighboring pairs of suffix histories of length $T - 1$. For

$$\lambda = (z^{(0)}[1..T-1], z^{(1)}[1..T-1]) \in \Lambda,$$

let $V_\lambda^{(b)}$ be the distribution of the length- $(T - 1)$ transcript generated by $\mathcal{M}^{(c)}$ on the fixed suffix input $z^{(b)}[1..T-1]$. Define

$$g_\lambda := \text{Pow}(V_\lambda^{(0)} \parallel V_\lambda^{(1)}) \quad \text{and} \quad g := \sup_{\lambda \in \Lambda} g_\lambda.$$

By definition of the supremum of power functions, g is a power function.

We claim that $\mathcal{M}^{(c)}$ is non-adaptively g -DP up to horizon $T - 1$. For deterministic non-adaptive suffix adversaries this is immediate from the definition of g . Randomized non-adaptive suffix adversaries are mixtures of deterministic ones, and the same bound follows from Fact A.9.

Moreover, $g \leq f$. To see this, fix any $\lambda \in \Lambda$ and consider the non-adaptive length- T adversary for $\mathcal{M}$ that submits the common first input c in both branches and then submits the suffix pair λ . The first-round output has the same distribution under the two secret bits, and the remaining transcript has law $V_\lambda^{(b)}$ under bit b . Hence the power function of this full non-adaptive transcript pair is

$$\text{Id} \otimes g_\lambda = g_\lambda.$$

The non-adaptive f -DP assumption for $\mathcal{M}$ gives $g_\lambda \leq f$ for every $\lambda \in \Lambda$. Taking the supremum over λ gives $g \leq f$.

Now condition on a possible first-round output u in the support of $U^{(0)} = U^{(1)}$. After observing u , the original deterministic adaptive adversary $\mathcal{A}$ induces a deterministic adaptive suffix adversary $\mathcal{A}_u$ for the shifted mechanism $\mathcal{M}^{(c)}$: it continues exactly as $\mathcal{A}$ would continue after the first transcript symbol u . Let

$$W^{(b)}(u) := \text{view}(\mathcal{A}_u \leftrightarrow (\mathcal{M}^{(c)})_b^{\text{pair}})$$

be the distribution of the remaining transcript under secret bit b , and write

$$h_u := \text{Pow}(W^{(0)}(u) \parallel W^{(1)}(u)).$$

Since $\mathcal{M}^{(c)}$ is quantitatively decomposable and non-adaptively g -DP up to horizon $T - 1$, the induction hypothesis gives

$$h_u \leq g$$

for every such first-round output u .

Finally, the full transcript under secret bit b has the same distribution as

$$(U, W^{(b)}(U)),$$

where $U \sim U^{(0)} = U^{(1)}$. The first coordinate has identical distribution under the two secret bits, and, conditional on $U = u$, the continuation distributions have power function at most g . Applying Fact A.10 with first-stage profile Id and second-stage profile g yields

$$\text{Pow}\left(\text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_0^{\text{pair}}) \parallel \text{view}(\mathcal{A} \leftrightarrow \mathcal{M}_1^{\text{pair}})\right) \leq \text{Id} \otimes g = g \leq f.$$

This proves Case 2.

The two cases complete the induction step. Hence the finite-horizon adaptive f -DP guarantee holds for every deterministic adaptive adversary. As argued at the beginning, joint convexity then extends the same guarantee to randomized adaptive adversaries. Therefore $\mathcal{M}$ is adaptively f -DP for every finite horizon. $\square$

Remark E.5 (Vector-Valued Aggregates). Definition E.3 states quantitative decomposability using a scalar distance parameter. This is sufficient for the theorem as used in the main text. Some standard vector-valued mechanisms, however, have an exact one-round power profile that depends on the full difference vector rather than only on its norm. For example, for coordinate-wise Laplace noise on $\mathbb{R}^k$, the exact profile of $u + \eta$ versus $u' + \eta$ may depend on the vector $u - u'$.

The theorem and proof extend verbatim to this setting by replacing the scalar distance $d(u, u')$ with an invariant difference map $\Delta(u, u')$ taking values in some parameter space, such as $\Delta(u, u') = u - u' \in \mathbb{R}^k$, and replacing $g_t(\theta)$ by a family $g_t(v)$ indexed by this difference parameter. The only property needed in the proof is invariance under common aggregation context:

$$\Delta(a \oplus u \oplus b, a \oplus u' \oplus b) = \Delta(u, u').$$

Under this vector-indexed variant, the binary-tree prefix-sum mechanism with coordinate-wise Laplace noise is covered exactly for vector updates in $\mathbb{R}^k$. In the main text, we state the simpler scalar version because the usual (ε, δ) consequences can also be obtained by upper-bounding the vector-indexed profile in terms of a norm, such as $\|u - u'\|_1$ for coordinate-wise Laplace noise.

E.1. Why Independent Decomposability Alone Is Insufficient

The certification theorem in Section E requires quantitative decomposability, not merely independent decomposability. The following example shows why the stronger condition is needed. Even when each round uses fresh independent randomness and the round- t output is a deterministic function of the input prefix and the fresh seed, a non-adaptive approximate-DP guarantee need not lift to adaptive adversaries. The obstruction is that the one-round leakage can depend on the adaptive common context, rather than only on the location and magnitude of the Hamming discrepancy.

Counterexample with Fresh Independent Randomness. Consider a two-round mechanism with input and output alphabets $\{0, 1\}$. Write $x_1 = b$ and $x_2 = q$. Using independent fresh randomness in the two rounds, the mechanism outputs $Y_1, Y_2 \in \{0, 1\}$ with

$$\Pr[Y_1 = 1 \mid b = 0] = 0.68, \quad \Pr[Y_1 = 1 \mid b = 1] = 0.65,$$

and

	$q = 0$	$q = 1$
$b = 0$	$\Pr[Y_2 = 1] = 0.28$	$\Pr[Y_2 = 1] = 0.20$
$b = 1$	$\Pr[Y_2 = 1] = 0.19$	$\Pr[Y_2 = 1] = 0.28$

Since each output is a deterministic function of the current prefix and a fresh independent seed, the mechanism is independently decomposable in the sense of Definition E.1.

For every fixed input stream (b, q) , the transcript distribution is the product distribution of Y_1 and Y_2 . Since two-sided $(0, \delta)$ -DP is equivalent to total variation distance at most δ , it suffices to check the four Hamming-neighboring pairs:

Fixed neighboring streams	Total variation distance
$(0, 0)$ and $(1, 0)$	0.09
$(0, 1)$ and $(1, 1)$	0.08
$(0, 0)$ and $(0, 1)$	0.08
$(1, 0)$ and $(1, 1)$	0.09.

Hence the mechanism is non-adaptively $(0, 0.09)$ -DP.

Now consider the deterministic adaptive adversary that chooses the challenge pair $x_1^{(0)} = 0$ and $x_1^{(1)} = 1$ at round 1, and, after observing $Y_1 = y$, submits the common second input $x_2^{(0)} = x_2^{(1)} = y$. This is a valid adaptive Hamming-neighbor adversary: the two submitted streams differ only at coordinate 1, but their common second input depends on the first release.

The resulting transcript distributions are:

(Y_1, Y_2)	World 0	World 1
(0, 0)	0.2304	0.2835
(0, 1)	0.0896	0.0665
(1, 0)	0.5440	0.4680
(1, 1)	0.1360	0.1820

For $S = \{(0, 1), (1, 0)\}$, we have

$$\Pr_0[S] - \Pr_1[S] = 0.6336 - 0.5345 = 0.0991 > 0.09.$$

Therefore,

$$\text{TV}(\text{View}_0, \text{View}_1) = 0.0991,$$

so the mechanism is not adaptively $(0, 0.09)$ -DP.

This does not contradict Theorem E.4, because the theorem assumes quantitative decomposability. The example shows that fresh independent randomness alone is not enough: the adaptive adversary can choose a common later input that changes how the earlier Hamming discrepancy is revealed. Quantitative decomposability rules out precisely this behavior by forcing the one-round power function to depend only on the distance between the relevant aggregates, with common aggregation context removed by the invariant-pair property.

F. Streaming Prefix-Sum Mechanism

There have been numerous works on differentially private prefix-sum algorithms since the *binary tree mechanism* was introduced (Chan et al., 2010; 2011). We first state the standard pure-DP primitive with a simultaneous high-probability error bound, and then derive the adaptive, always-error-bounded, and consistent variant needed by RandBin.

Problem Setting. At each time step $t \geq 1$, an integer x_t is passed to an interactive mechanism, which releases an estimate S_t of the prefix sum

$$Q_t := \sum_{\tau=1}^t x_\tau.$$

The additive error at time t is $|S_t - Q_t|$, and the adversary may choose later inputs after observing earlier releases.

Neighboring Notion. Two equal-length input streams are neighboring if they differ in at most one time step, where the two values at that step differ by exactly 1. As aforementioned, an unbounded interactive mechanism is adaptively (ε, δ) -DP if its truncation to every finite horizon $T > 0$ is adaptively (ε, δ) -DP.

As in (Chan et al., 2022), we begin with a pure-DP prefix-sum mechanism having a simultaneous high-probability error bound.

Fact F.1 (Pure-DP Prefix Sums (Chan et al., 2011)). *For every $\varepsilon > 0$, there exists an integer-valued interactive mechanism $\text{HybridStreamSum}_\varepsilon$, defined on streams in $\mathbb{Z}$, with the following properties.*

- **Non-adaptive privacy.** *For the unit-Hamming neighboring notion above, $\text{HybridStreamSum}_\varepsilon$ is non-adaptively $(\varepsilon, 0)$ -DP. This guarantee holds for the stronger transcript consisting of all noisy interval counts described below.*
- **Simultaneous error.** *For every $\beta \in (0, 1)$, there is a deterministic nondecreasing function*

$$B_{\varepsilon, \beta}(t) = O\left(\frac{1}{\varepsilon} \log(t+1) \left(\log(t+1) + \log \frac{1}{\beta}\right)\right)$$

such that, with probability at least $1 - \beta$,

$$\left| Y_t - \sum_{\tau=1}^t x_\tau \right| \leq B_{\varepsilon, \beta}(t) \quad \text{for every } t \geq 1,$$

where Y_t is the estimate released at time t . The simultaneous success event may be chosen to depend only on the mechanism's noises. Consequently, the same event applies when the inputs are selected adaptively from previous releases.

- **Certified core structure.** The mechanism admits a p -sum implementation whose stronger core transcript contains exactly one new noisy interval count in each original round. The round- t core output can be written as

$$Z_t = \sum_{i \in I_t} x_i + \eta_t, \quad I_t = [c_t..t] \subseteq [1..t].$$

The schedule $(I_t)_{t \geq 1}$ is data-independent, and the noises η_t are fresh and independent across rounds. They may be taken to be symmetric geometric noises. The exact one-round power function depends only on the absolute integer shift between the two interval sums and is nondecreasing in that shift. The published estimate Y_t is a deterministic function of the core releases available by time t .

The stronger-transcript statement follows from the standard p -sum view of the hybrid mechanism. For completeness, its simultaneous infinite-horizon error guarantee can be obtained as follows. For time t , apply the standard sub-exponential tail bound to the sum of the relevant geometric noises with failure probability

$$\beta_t := \frac{\beta}{t(t+1)}.$$

Since $\sum_{t \geq 1} \beta_t = \beta$ and

$$\log \frac{1}{\beta_t} = O\left(\log(t+1) + \log \frac{1}{\beta}\right),$$

a union bound gives the stated error bound simultaneously for all $t \geq 1$. Moreover, after subtracting the true prefix sum, the error is a deterministic function only of the core noises. Replacing the usual Laplace perturbations by their symmetric geometric analogues preserves pure DP and the same tail order while making all releases integer-valued.

Theorem F.2 (Adaptive Prefix Sums with Always-Valid Error and Consistency). *For every $\varepsilon > 0$, $\delta \in (0, 1)$, and integer interval $[a..b] \subseteq \mathbb{Z}$, there exists an interactive mechanism $\text{StreamSum}_{\varepsilon, \delta}$, defined on all integer streams, that is adaptively (ε, δ) -DP for unit-Hamming neighboring streams.*

Moreover, there is a deterministic nondecreasing integer-valued function

$$E_{\varepsilon, \delta} : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}, \quad E_{\varepsilon, \delta}(0) = 0,$$

such that, for every $t \geq 1$,

$$E_{\varepsilon, \delta}(t) = O\left(1 + \frac{1}{\varepsilon} \log(t+1) \left(\log(t+1) + \log \frac{1}{\delta}\right)\right).$$

On every execution whose inputs satisfy $x_t \in [a..b]$, the integer outputs S_t satisfy, simultaneously for every $t \geq 1$,

$$\begin{aligned} \left| S_t - \sum_{\tau=1}^t x_\tau \right| &\leq E_{\varepsilon, \delta}(t), & (\text{always-valid error}) \\ S_t - S_{t-1} &\in [a..b], & (\text{consistency}) \end{aligned}$$

where $S_0 := 0$. These properties hold for every realization of the mechanism's randomness.

Proof. Let $\text{HybridStreamSum}_\varepsilon$ be the mechanism in Fact F.1, and abbreviate

$$B(t) := B_{\varepsilon, \delta}(t).$$

Define

$$E_{\varepsilon, \delta}(0) := 0, \quad E_{\varepsilon, \delta}(t) := \lceil B(t) \rceil + 2 \quad (t \geq 1).$$

Because B is nondecreasing, $E_{\varepsilon, \delta}$ is a nondecreasing integer-valued function with the asymptotic bound stated in the theorem. We construct the claimed mechanism in three stages.

1. *Adaptive privacy of the raw mechanism.* Use the stronger core transcript from Fact F.1. Its round- t release is

$$Z_t = \sum_{\tau \in I_t} x_\tau + \eta_t,$$

where $I_t \subseteq [1..t]$ is data-independent and η_t is fresh and independent of all other rounds.

Take the aggregation operation to be integer addition and define

$$d(u, u') := |u - u'|.$$

Then $(d, +)$ is an invariant pair because, for all integers a, b, u, u' ,

$$d(a + u + b, a + u' + b) = |u - u'| = d(u, u').$$

For every nonnegative integer k , define

$$\tilde{g}_t(k) := \text{Pow}(\eta_t \| \eta_t + k).$$

By symmetry, the same power function is obtained for a shift by $-k$. Moreover, for symmetric geometric noise, $\tilde{g}_t(k)$ is nondecreasing in k . Extend it to $\theta \geq 0$ by

$$g_t(\theta) := \tilde{g}_t(\lfloor \theta \rfloor).$$

Because every relevant distance is an integer, this extension preserves the required exact identities.

Translation invariance and symmetry now give, for every pair of integer input histories,

$$\text{Pow} \left(\sum_{\tau \in I_t} x_\tau + \eta_t \left\| \sum_{\tau \in I_t} x'_\tau + \eta_t \right. \right) = g_t \left(\left| \sum_{\tau \in I_t} x_\tau - \sum_{\tau \in I_t} x'_\tau \right| \right).$$

Thus the noisy-count core is quantitatively decomposable with respect to the invariant pair $(d, +)$ in the sense of Definition E.3. By Fact F.1, this core is non-adaptively $(\varepsilon, 0)$ -DP for every finite horizon. Therefore, Theorem E.4 implies that it is adaptively $(\varepsilon, 0)$ -DP for every finite horizon. Since the raw prefix estimates are causal post-processing of the core transcript, $\text{HybridStreamSum}_\varepsilon$ is also adaptively $(\varepsilon, 0)$ -DP.

2. *An always-valid error bound.* Given the raw candidate Y_t at time t , define the clipped release

$$\hat{Y}_t := \text{clip}(Y_t, [Q_t - E_{\varepsilon, \delta}(t), Q_t + E_{\varepsilon, \delta}(t)]),$$

where

$$\text{clip}(y, [\ell, u]) := \min\{\max\{y, \ell\}, u\}.$$

This immediately gives $|\hat{Y}_t - Q_t| \leq E_{\varepsilon, \delta}(t)$ on every execution. Because the clipping interval depends on the private prefix sum, however, adaptive privacy does not follow from ordinary post-processing. We prove it directly.

Fix a finite horizon T and, by joint convexity, a deterministic adaptive adversary $\mathcal{A}$. For every candidate released transcript $v[1..T]$, let

$$x_t^{(c)}(v[1..t-1]), \quad c \in \{0, 1\},$$

be the c -th input selected by $\mathcal{A}$ at time t after observing $v[1..t-1]$, and define the counterfactual prefix sums

$$Q_t^{(c)}(v[1..t-1]) := \sum_{\tau=1}^t x_\tau^{(c)}(v[1..\tau-1]).$$

We may assume that the adversary generates unit-Hamming neighboring streams after every transcript prefix; its behavior after a prefix at which the paired simulation would terminate is immaterial and can be redefined arbitrarily. Hence

$$\left| Q_t^{(0)}(v[1..t-1]) - Q_t^{(1)}(v[1..t-1]) \right| \leq 1 \tag{4}$$

for every transcript prefix and every $t \leq T$.

Let P_c and $\hat{P}_c$ be the laws of the raw and clipped released transcripts, respectively, in branch c . Define the common measurable set

$$G_T := \left\{ v[1..T] : \left| v_t - Q_t^{(c)}(v[1..t-1]) \right| < E_{\varepsilon, \delta}(t) \text{ for every } t \leq T \text{ and } c \in \{0, 1\} \right\}. \quad (5)$$

This is one fixed subset of the transcript space, defined using both counterfactual input sequences.

For branch c , let

$$H_c := \left\{ \left| Y_t - Q_t^{(c)}(Y[1..t-1]) \right| \leq B(t) \text{ for every } t \leq T \right\}.$$

The noise-only simultaneous error event in Fact F.1 gives $P_c(H_c) \geq 1 - \delta$, even for adaptively selected inputs. On H_c , (4) implies, for every $t \leq T$,

$$\begin{aligned} \left| Y_t - Q_t^{(c)}(Y[1..t-1]) \right| &\leq B(t) < E_{\varepsilon, \delta}(t), \\ \left| Y_t - Q_t^{(1-c)}(Y[1..t-1]) \right| &\leq B(t) + 1 < E_{\varepsilon, \delta}(t). \end{aligned}$$

Hence $H_c \subseteq G_T$ and

$$P_c(G_T^c) \leq \delta. \quad (6)$$

Couple the raw and clipped executions in a fixed branch c using the same complete noise sequence. We claim that

$$Y[1..T] \in G_T \iff \hat{Y}[1..T] \in G_T, \quad (7)$$

and that the two transcripts are identical whenever either condition holds. For the forward direction, induct on t . If $Y[1..t-1] = \hat{Y}[1..t-1]$, then the adversary supplies the same input and the shared noises give the same raw candidate in both executions. Because $Y[1..T] \in G_T$, that candidate lies strictly inside the branch- c clipping interval, so clipping leaves it unchanged. For the reverse direction, the same induction applies: if $\hat{Y}[1..T] \in G_T$, then $\hat{Y}_t$ lies strictly inside the clipping interval; if clipping had changed the raw candidate, the released value would instead equal an endpoint. Thus clipping did not occur.

It follows from (7) that, for every measurable transcript event A ,

$$\hat{P}_c(A \cap G_T) = P_c(A \cap G_T), \quad \hat{P}_c(G_T^c) = P_c(G_T^c) \leq \delta. \quad (8)$$

Using adaptive pure DP of the raw mechanism, we obtain

$$\begin{aligned} \hat{P}_0(A) &\leq \hat{P}_0(A \cap G_T) + \hat{P}_0(G_T^c) \\ &= P_0(A \cap G_T) + P_0(G_T^c) \\ &\leq e^\varepsilon P_1(A \cap G_T) + \delta \\ &= e^\varepsilon \hat{P}_1(A \cap G_T) + \delta \\ &\leq e^\varepsilon \hat{P}_1(A) + \delta. \end{aligned}$$

The same argument with the branches interchanged gives the reverse inequality. Thus the clipped mechanism is adaptively (ε, δ) -DP for every finite horizon T , and hence for the unbounded interaction.

3. *Consistency.* Apply the following deterministic online projection to the clipped transcript:

$$S_0 := 0, \quad S_t := \text{clip} \left(\hat{Y}_t, [S_{t-1} + a, S_{t-1} + b] \right).$$

This is causal post-processing, so it preserves adaptive (ε, δ) -DP: any adversary interacting with the projected releases, together with this causal wrapper, induces an adversary for the clipped mechanism. The projection immediately gives $S_t - S_{t-1} \in [a..b]$.

It remains to verify the error. Suppose inductively that $|S_{t-1} - Q_{t-1}| \leq E_{\varepsilon, \delta}(t-1)$. Since $x_t \in [a..b]$ and $E_{\varepsilon, \delta}$ is nondecreasing,

$$\text{dist}(Q_t, [S_{t-1} + a, S_{t-1} + b]) \leq |Q_{t-1} - S_{t-1}| \leq E_{\varepsilon, \delta}(t).$$

Projection onto an interval satisfies

$$|\text{clip}(y, J) - q| \leq \max\{|y - q|, \text{dist}(q, J)\}.$$

Together with $|\widehat{Y}_t - Q_t| \leq E_{\varepsilon, \delta}(t)$, this yields $|S_t - Q_t| \leq E_{\varepsilon, \delta}(t)$. The induction starts from $S_0 = Q_0 = 0$, proving both claimed execution-wise properties. $\square$

F.1. Prefix Sums via Lower-Triangular Matrix Factorizations

The mechanism above is the integer prefix-sum scheduler used internally by RandBin. We now separately analyze the Gaussian lower-triangular aggregators used in the experiments. These aggregators require adaptive privacy, but not the probability-one error or consistency properties of Theorem F.2.

Fix a horizon T and write a scalar stream as $x \in \mathbb{R}^T$. The prefix sums form the linear workload

$$M_{\text{count}} x \in \mathbb{R}^T, \quad M_{\text{count}}[t, i] = \mathbf{1}[t \geq i],$$

$$\text{so } (M_{\text{count}} x)_t = \sum_{i=1}^t x_i.$$

We use event-level neighbors $x \sim x'$ satisfying

$$x - x' = \xi e_i \quad \text{for some } i \in [T] \text{ and } |\xi| \leq \Delta.$$

For scalar unit-count streams, $\Delta = 1$. For vector-valued updates, the two streams differ in at most one time block, the ℓ_2 -norm of that block difference is at most Δ , and the corresponding linear operator is $R \otimes I$.

One-hop versus two-hop accounting. Theorem F.2 is used for RandBin's internal scheduler under the unit-Hamming notion stated above. The privacy bounds in this subsection are likewise primitive-level, distance-one guarantees. When a Gaussian prefix-sum mechanism is used as the downstream aggregator, however, a single input edit may induce at most two adjacent Hamming changes at RandBin's emitted-update interface (Lemma 4.1). Hence adaptive group privacy (Fact B.6) gives

$$(2\widehat{\varepsilon}_a, (1 + e^{\widehat{\varepsilon}_a})\widehat{\delta}_a)\text{-DP}.$$

We therefore set

$$\widehat{\varepsilon}_a = \frac{\varepsilon_a}{2}, \quad \widehat{\delta}_a = \frac{\delta_a}{1 + e^{\varepsilon_a/2}},$$

so that the total downstream cost is at most $(\varepsilon_a, \delta_a)$.

Privacy Notions: Rényi DP (RDP) and zCDP. We first state the usual fixed-stream, non-adaptive guarantees; their adaptive lift is established below. Let P, Q have densities p, q with respect to a common base measure. For $\alpha > 1$,

$$D_\alpha(P \| Q) := \frac{1}{\alpha - 1} \log \int p(y)^\alpha q(y)^{1-\alpha} dy.$$

A fixed-stream mechanism $\mathcal{M}$ is $(\alpha, \varepsilon_\alpha)$ -RDP if, for all neighboring $x \sim x'$,

$$D_\alpha(\mathcal{M}(x) \| \mathcal{M}(x')) \leq \varepsilon_\alpha.$$

It is ρ -zCDP if, for every $\alpha > 1$ and every neighboring $x \sim x'$,

$$D_\alpha(\mathcal{M}(x) \| \mathcal{M}(x')) \leq \rho\alpha.$$

Equivalently, ρ -zCDP implies $(\alpha, \rho\alpha)$ -RDP simultaneously for every $\alpha > 1$.

Converting RDP to (ε, δ) -DP. If $\mathcal{M}$ satisfies $(\alpha, \varepsilon_\alpha)$ -RDP for some $\alpha > 1$, then, for every $\delta \in (0, 1)$, it satisfies (ε, δ) -DP with

$$\varepsilon = \varepsilon_\alpha + \frac{\log(1/\delta)}{\alpha - 1}.$$

Optimizing over the order gives

$$\varepsilon(\delta) = \inf_{\alpha > 1} \left\{ \varepsilon_\alpha + \frac{\log(1/\delta)}{\alpha - 1} \right\}.$$

Lower-Triangular Factorization Mechanism. Let $M_{\text{count}} = LR$ be a possibly rectangular factorization, where $R \in \mathbb{R}^{m \times T}$ and $L \in \mathbb{R}^{T \times m}$ are lower-triangular in time, so all core releases and published outputs are causal. Define the column sensitivity

$$s(R) := \max_{i \in [T]} \|Re_i\|_2 = \|R\|_{1 \rightarrow 2}.$$

Consider

$$\mathcal{M}_{L,R,\sigma}(x) := L(Rx + z), \quad z \sim \mathcal{N}(0, \sigma^2 I_m),$$

where the coordinates of z are independent. Since multiplication by L is post-processing, privacy is governed by the Gaussian core $Rx + z$.

Lemma F.3 (zCDP/RDP Bounds for $L(Rx + z)$). *For neighboring $x \sim x'$ differing at coordinate i ,*

$$\|R(x - x')\|_2 = |\xi| \|Re_i\|_2 \leq \Delta s(R).$$

Hence, in the fixed-stream experiment, $\mathcal{M}_{L,R,\sigma}$ satisfies ρ -zCDP with

$$\rho = \frac{\Delta^2 s(R)^2}{2\sigma^2},$$

and, for every $\alpha > 1$, it satisfies $(\alpha, \varepsilon_\alpha)$ -RDP with

$$\varepsilon_\alpha = \alpha \rho = \frac{\alpha \Delta^2 s(R)^2}{2\sigma^2}.$$

Binary Tree Mechanism as a Factorization. Assume first that $T = 2^h$ for an integer $h \geq 0$; otherwise, pad to the next power of two. Index the complete binary-tree nodes by their dyadic intervals. Let $q = R_{\text{tree}}x$ be the vector of all dyadic-interval sums, where R_{tree} is the corresponding 0/1 incidence matrix. Each leaf belongs to exactly its $h + 1$ ancestors, and each prefix $[1..t]$ has a canonical dyadic partition of size at most $h + 1$. Therefore, there is a causal reconstruction matrix L_{tree} such that $M_{\text{count}} = L_{\text{tree}}R_{\text{tree}}$.

Proposition F.4 (Privacy Bounds for the Binary-Tree Factorization). *For $T = 2^h$,*

$$s(R_{\text{tree}})^2 = \|R_{\text{tree}}\|_{1 \rightarrow 2}^2 = h + 1.$$

Consequently, the Gaussian tree mechanism $L_{\text{tree}}(R_{\text{tree}}x + z)$, with $z \sim \mathcal{N}(0, \sigma^2 I_m)$, is non-adaptively ρ_{tree} -zCDP with

$$\rho_{\text{tree}} = \frac{\Delta^2(h + 1)}{2\sigma^2},$$

and, for every $\alpha > 1$, is $(\alpha, \varepsilon_{\alpha, \text{tree}})$ -RDP with

$$\varepsilon_{\alpha, \text{tree}} = \frac{\alpha \Delta^2(h + 1)}{2\sigma^2}.$$

For general T , the same bounds hold with $h := \lceil \log_2 T \rceil$.

A Smooth Lower-Triangular Factorization (Fichtenberger et al., 2023). Fichtenberger et al. give an explicit Toeplitz lower-triangular factorization $M_{\text{count}} = LR$ with $L = R$, defined by

$$f(0) = 1, \quad f(k) = \left(\frac{2k-1}{2k} \right) f(k-1) \quad (k \geq 1), \quad L[t, i] = R[t, i] = f(t-i) \quad (t \geq i),$$

and zero otherwise. Let γ be the Euler–Mascheroni constant and define

$$\Psi(T) := 1 - \frac{1-\gamma}{\pi} + \frac{\ln T}{\pi} + \frac{2}{T}.$$

Proposition F.5 (Privacy Bounds for the Fichtenberger–Henzinger–Upadhyay Factorization). *For the factorization above,*

$$s(R)^2 = \|R\|_{1 \rightarrow 2}^2 = \|L\|_{2 \rightarrow \infty}^2 \leq \Psi(T).$$

Consequently, $L(Rx + z)$ with $z \sim \mathcal{N}(0, \sigma^2 I_T)$ is non-adaptively ρ_{FHU} -zCDP with

$$\rho_{\text{FHU}} = \frac{\Delta^2 \|R\|_{1 \rightarrow 2}^2}{2\sigma^2} \leq \frac{\Delta^2 \Psi(T)}{2\sigma^2},$$

and, for every $\alpha > 1$, is $(\alpha, \varepsilon_{\alpha, \text{FHU}})$ -RDP with

$$\varepsilon_{\alpha, \text{FHU}} = \frac{\alpha \Delta^2 \|R\|_{1 \rightarrow 2}^2}{2\sigma^2} \leq \frac{\alpha \Delta^2 \Psi(T)}{2\sigma^2}.$$

Adaptive-Safety Audit. For each row $r_j^\top$ of R , let $\tau(j)$ be its public release time, so that

$$\text{supp}(r_j) \subseteq [1.. \tau(j)].$$

Let

$$J_t := \{j : \tau(j) = t\},$$

and let A_t be the matrix whose rows are $\{r_j^\top : j \in J_t\}$. Package all rows released at the same original input time into the single vector-valued core release

$$Z_t = A_t x + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2 I_{|J_t|}).$$

If $J_t = \emptyset$, take Z_t to be a fixed data-independent dummy release. The noise vectors η_t are fresh and independent across original input times. Thus there is exactly one core output per round, as required by the interactive model.

To certify quantitative decomposability, represent the scalar update at coordinate i by the vector $x_i e_i \in \mathbb{R}^T$ and aggregate these vectors using addition. For vector-valued streams, use the analogous block vector $e_i \otimes x_i$.

In Definition E.3, take $I_t = [1..t]$. Because every row of A_t is supported on $[1..t]$,

$$A_t \left(\sum_{i \in I_t} x_i e_i \right) = A_t x.$$

For vector-valued streams, the same identity holds with $A_t \otimes I$.

Define the invariant difference map

$$D(u, u') := u' - u.$$

It satisfies

$$D(a + u + b, a + u' + b) = u' - u = D(u, u').$$

For a discrepancy vector v , define

$$g_t(v) := \text{Pow}(\eta_t \parallel \eta_t + A_t v).$$

Translation invariance gives the exact identity

$$\text{Pow}(A_t x + \eta_t \parallel A_t x' + \eta_t) = g_t(x' - x).$$

Thus the Gaussian core is quantitatively decomposable under the vector-indexed extension in Remark E.5. Since the rows released by time t depend only on $x[1..t]$, the core is causal, and its noise vectors are fresh and independent.

It remains to connect the lifted power-function guarantee to the stated RDP and zCDP bounds. Define

$$\mu := \frac{\Delta s(R)}{\sigma},$$

and let f_μ denote the power function of the one-dimensional Gaussian shift experiment

$$\mathcal{N}(0, 1) \quad \text{versus} \quad \mathcal{N}(\mu, 1).$$

For every fixed pair of neighboring streams,

$$\frac{\|R(x - x')\|_2}{\sigma} \leq \mu.$$

Consequently, the non-adaptive Gaussian-core experiment has power function at most f_μ . Theorem E.4 therefore gives the same f_μ upper bound for the transcript generated against every adaptive adversary.

If V_0, V_1 denote the resulting adaptive transcript laws, Gaussian experiment domination and data processing for Rényi divergence give, for every $\alpha > 1$,

$$D_\alpha(V_0 \| V_1) \leq \frac{\alpha \mu^2}{2} = \frac{\alpha \Delta^2 s(R)^2}{2\sigma^2}.$$

The same bound holds with the two branches interchanged. Hence the adaptive core satisfies the same

$$\rho = \frac{\Delta^2 s(R)^2}{2\sigma^2}$$

zCDP bound and the same $(\alpha, \alpha\rho)$ -RDP bounds as in the fixed-stream analysis. Since L is lower triangular, every published output is causal post-processing of the core releases available at that time. The same adaptive RDP/zCDP bounds, and their converted (ε, δ) -DP guarantees, therefore hold for the published mechanism.

Takeaway. Both Gaussian mechanisms fit the template $L(Rx + z)$, but have different column-sensitivity bounds: the binary tree has $s(R)^2 = \Theta(\log_2 T)$, while the smooth Toeplitz factorization satisfies

$$s(R)^2 \leq 1 - \frac{1 - \gamma}{\pi} + \frac{\ln T}{\pi} + \frac{2}{T}.$$

The adaptive-safety audit shows that these same privacy bounds hold under feedback-adaptive input selection.

G. Additional Experiment Details

G.1. Privacy Attack: Hamming-Style DP vs. Edit-Style DP in Continual Learning

Objective. We construct an attack scenario showing that a continual-learning mechanism based on prefix-sum aggregation can satisfy privacy under *Hamming-style adjacency* (record substitution), but fails under *edit-style adjacency* (single insertion/deletion with global shift). This demonstrates that Hamming-style DP guarantees do not automatically extend to edit-style (stream-level) privacy in continual-release settings.

Neighboring streams (edit adjacency). We construct two data streams D and D' over MNIST digits $\{0, 1\}$ with fixed batch size and fixed batching. Importantly, each occurrence of “0” or “1” corresponds to a *distinct training sample* from the dataset (not repeated copies of a single example); the symbols only denote the class labels.

The streams are constructed as follows:

- **Stream D .** All samples have label 0, except that the *first element of every odd-indexed batch* (batches 1, 3, 5, ...) is replaced by a sample with label 1.
- **Stream D' .** Obtained from D by deleting the first element, shifting the entire stream left by one position, and appending a 0-labeled sample at the end.

Thus D and D' differ by a *single edit operation* (delete+shift), and are neighbors under edit-style adjacency. However, the shift moves the injected “1” samples across batch boundaries, which produces *systematic differences in many minibatch compositions and gradients* across training rounds, despite the streams differing by only one edit.

Learning mechanism. We follow the framework of Algorithm 2 using a tree-based prefix-sum mechanism. Let $D_\star \in \{D, D'\}$ denote the *true private stream*. At round t , the t -th batch of $D_\star$ is fed into the current model $w^{(t-1)}$. Per-sample gradients are computed, clipped to norm G , and averaged to form the minibatch gradient $g_t^{(D_\star)} \in \mathbb{R}^d$. The update is made private using a continual prefix-sum mechanism: $S_t = \text{PrivStreamSum}(g_1^{(D_\star)}, \dots, g_t^{(D_\star)})$, and the model is updated by post-processing $w^{(t)} = w^{(0)} - \eta S_t$. All randomness is injected solely through the prefix-sum mechanism $\text{PrivStreamSum}(\cdot)$; gradient computation, clipping, and averaging are deterministic.

Algorithm 4 Streaming Majority-Vote Test

```

1: Input: Released private prefix sums  $S_t = \text{PrivStreamSum}(g_1^{(D^*)}, \dots, g_t^{(D^*)})$ ; candidate streams  $D, D'$ 
2: Output: Decision  $\hat{D}_t$  at each round  $t$ 
3: Initialize counters  $c_D \leftarrow 0$  and  $c_{D'} \leftarrow 0$ 
4: for  $t = 1, \dots, T$  do
5:   Observe noisy increment  $\Delta_t \leftarrow S_t - S_{t-1}$ 
6:   Using model  $w^{(t-1)}$ , compute candidate gradients:
7:     Feed batch  $t$  of  $D$  to obtain  $g_t^{(D)}$ 
8:     Feed batch  $t$  of  $D'$  to obtain  $g_t^{(D')}$ 
9:   if  $\|\Delta_t - g_t^{(D)}\| \leq \|\Delta_t - g_t^{(D')}\|$  then
10:      $c_D \leftarrow c_D + 1$ 
11:   else
12:      $c_{D'} \leftarrow c_{D'} + 1$ 
13:   end if
14:    $\hat{D}_t \leftarrow \arg \max\{c_D, c_{D'}\}$   $\triangleright$  majority over history
15: end for
    
```

Adversarial test. Assume the private stream is $D_* \in \{D, D'\}$. At each even-indexed round t , the adversary observes the released model and reconstructs the noisy prefix-sum increment

$$\Delta_t := S_t - S_{t-1}.$$

Using the current model $w^{(t-1)}$, the adversary then:

1. feeds the t -th batch of D into the model to compute the clipped-and-averaged gradient $g_t^{(D)}$;
2. feeds the t -th batch of D' into the model to compute the clipped-and-averaged gradient $g_t^{(D')}$.

The adversary compares the observed noisy increment Δ_t with the two candidate gradients $g_t^{(D)}$ and $g_t^{(D')}$, and casts a vote for the closer one in norm. Votes are aggregated over time using a majority rule.

Accuracy metric. For each round t , the test outputs a majority decision $\hat{D}_t$ using all evidence up to round t . The per-round accuracy is defined as $\text{Acc}(t) = \Pr \hat{D}_t = D_{\text{true}}$, and is estimated by averaging over multiple independent runs. This produces an accuracy curve as a function of training round.

Results. Figure 5 shows the accuracy curves for different noise levels in the tree-aggregation mechanism:

- For small noise ($\sigma \in [0.1, 1]$), the accuracy rapidly converges to 1, meaning the adversary almost surely distinguishes the two streams.
- For larger noise ($\sigma \in [2, 4]$), the accuracy decreases but remains *significantly above random guessing* (0.5).

Conclusion. This experiment provides a concrete separation: a continual prefix-sum mechanism that is secure under Hamming-style adjacency can leak information under edit-style adjacency. The failure is not due to local noise miscalibration, but to *temporal misalignment*: a single edit induces correlated shifts across many rounds, which can be aggregated by a streaming adversary. This shows that Hamming-style DP does not imply edit-style DP for continual-release learning systems.

G.2. Experiments Investigating Properties of RandBin

Batch-size effects. The mean batch-size is $\frac{3U}{2} = \Theta(\frac{1}{\epsilon} \log \frac{1}{\delta})$. Larger ϵ and δ lead to smaller batch sizes and more frequent updates; see Figure 6 for typical values of (ϵ, δ) .

Backlog validation. We validate that the backlog of pending users remains small in practice. We add a normalized-backlog plot, namely Q_t/t , versus time for the same settings as the raw-backlog plot in Figure 7. This directly shows that while

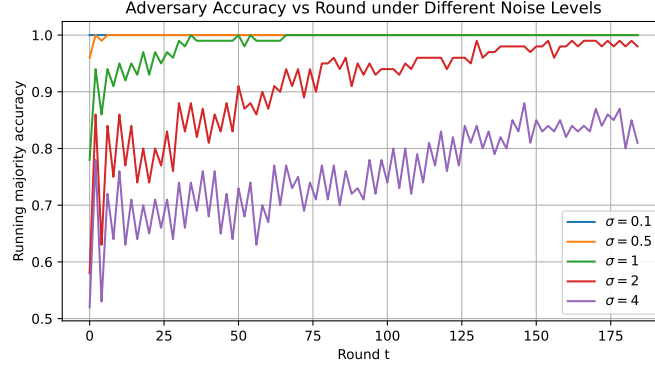


Figure 5. Adversary accuracy over rounds for different noise levels σ .

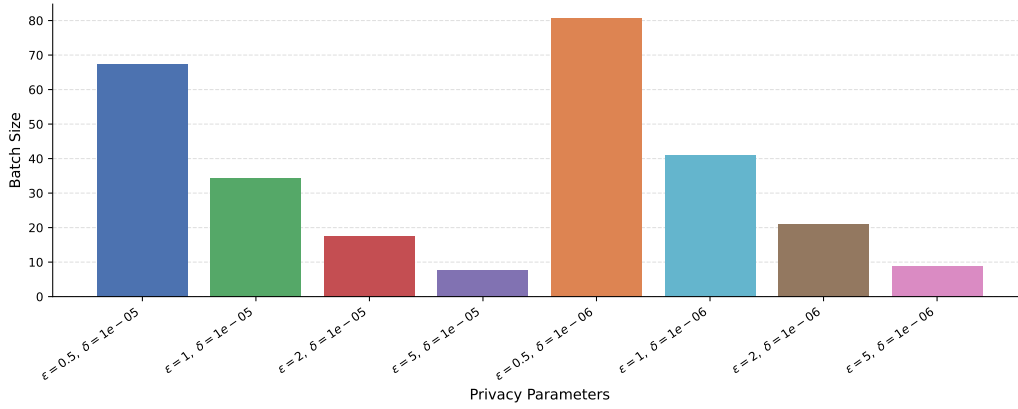


Figure 6. Mean batch sizes under different privacy parameters.

the raw backlog can grow with the horizon, the fraction of pending arrivals becomes negligible over time. This is the practically relevant view in light of Lemma 4.2, because it shows that the fraction of events currently pending becomes asymptotically negligible even when the raw backlog is horizon-dependent.

G.3. Deep-Network Experiments for CIFAR10 and MNIST

We provide additional experimental results for CIFAR-10 and MNIST datasets in Figure 8, complementing the EMNIST results in Figure 3.

G.4. Experiments for Application of Our Framework to ADMM

ADMM synthetic experiment overview. We include an ADMM experiment as a representative, more general optimizer that still fits our gradient-based update model. We instantiate Algorithm 2 with RandBin and PrivStreamSum under the single-edit stream definition, using the linearized ADMM formulation of Chan et al. (2024). Below we provide the background updates and the synthetic setup and results.

ADMM Background and Private Updates. We follow the description from (Chan et al., 2024). Let n, m, ℓ be positive integers, $F : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^\ell \rightarrow \mathbb{R}$ be convex, $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times \ell}$ be matrices, and $c \in \mathbb{R}^m$ a vector. ADMM solves $\min_{x,y} F(x) + g(y)$ subject to $Ax + By = c$ with dual $\lambda \in \mathbb{R}^m$ and penalty $\beta > 0$.

Streaming setting. We access F through a distribution $\mathcal{D}$ of functions with $\mathbf{E}_{f \leftarrow \mathcal{D}}[f(x)] = F(x)$. A stream of users $(f_t : t \geq 1)$ arrives, each f_t sampled independently from $\mathcal{D}$.

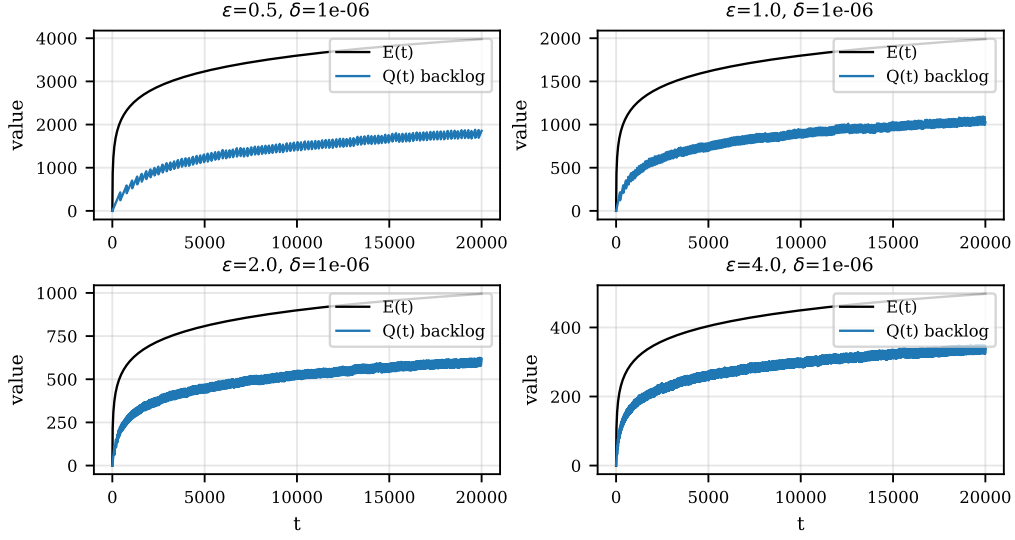
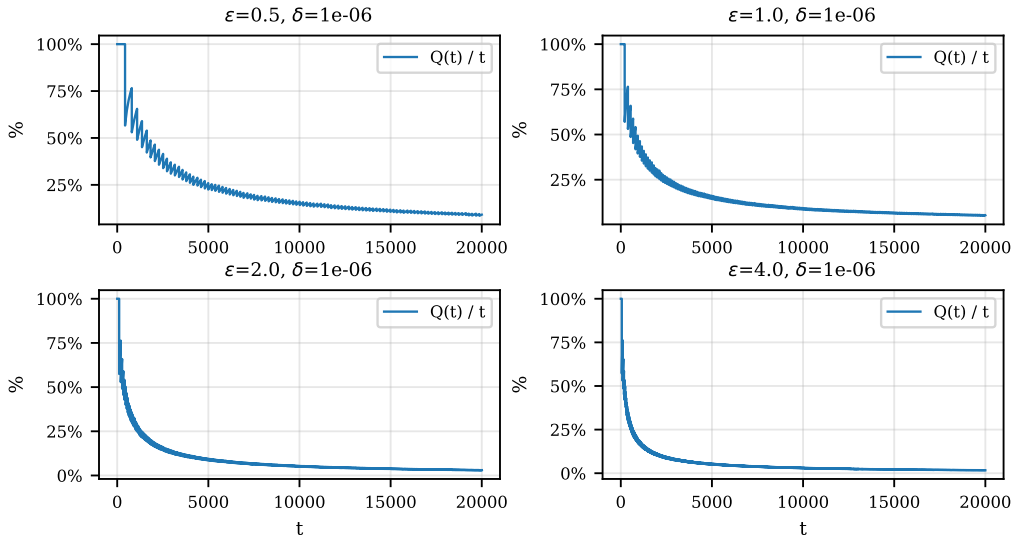

 (a) Raw backlog Q_t over time.

 (b) Normalized backlog Q_t/t for the same runs.

Figure 7. Privacy-implied latency under different privacy parameters. While the raw backlog may increase with the horizon, the pending fraction decreases over time and remains small at the end of training.

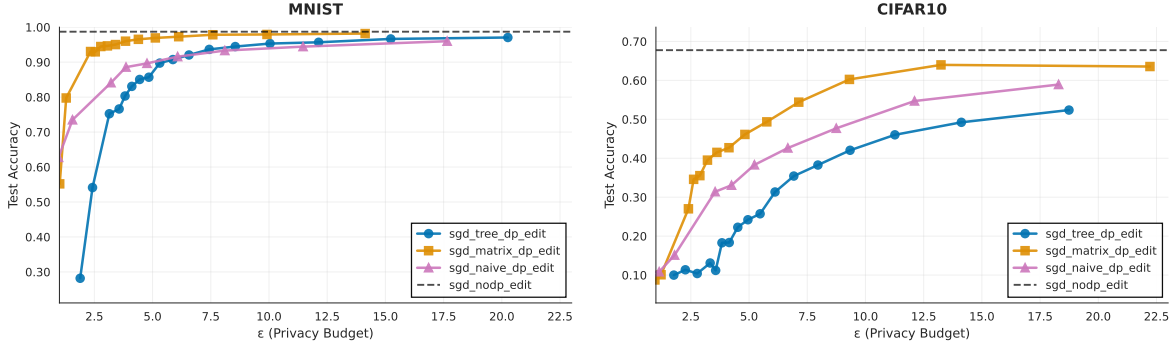


Figure 8. Accuracy-privacy tradeoff on MNIST and CIFAR-10 with single-edit privacy. All algorithms use the RandBin wrapper (sgdt_naive_dp_edit, sgdt_tree_dp_edit, and sgdt_matrix_dp_edit).

Algorithm 5 One ADMM Iteration

- 1: **Input:** Previous $(x_{t-1}, \lambda_{t-1}) \in \mathbb{R}^n \times \mathbb{R}^m$ and function $f_t : \mathbb{R}^n \rightarrow \mathbb{R}$
 - 2: **Output:** (x_t, λ_t)
 - 3: $y_{t-1} \leftarrow \mathcal{G}(x_{t-1}, \lambda_{t-1})$
 - 4: $\lambda_t \leftarrow \lambda_{t-1} - \beta(Ax_{t-1} + By_{t-1} - c)$
 - 5: $x_t \leftarrow \mathcal{F}^{f_t}(x_{t-1}, y_{t-1}, \lambda_t)$ $\triangleright$ oracle access to $\nabla f_t(\cdot)$
 - 6: **return** (x_t, λ_t)
-

One ADMM iteration. Assume each f is $\frac{1}{\eta}$ -smooth. For a fixed f and linearization point $\hat{x}$, define

$$\mathcal{L}_{\hat{x}}^f(x, y, \lambda) := f(\hat{x}) + \langle \nabla f(\hat{x}), x - \hat{x} \rangle + \mathcal{H}(x, y, \lambda) + \frac{1}{2\eta} \|x - \hat{x}\|^2$$

and

$$\mathcal{H}(x, y, \lambda) := g(y) - \langle \lambda, Ax + By - c \rangle + \frac{\beta}{2} \|Ax + By - c\|^2.$$

Let $\mathcal{G}(x, \lambda) := \arg \min_y \mathcal{H}(x, y, \lambda)$ and

$$\mathcal{F}^f(\hat{x}, y, \lambda) := \arg \min_x \mathcal{L}_{\hat{x}}^f(x, y, \lambda) = M^{-1} \{ \hat{x} - \eta \cdot [\nabla f(\hat{x}) + E(y, \lambda)] \},$$

where $M = \mathbb{I} + \eta\beta A^\top A$ and $E(y, \lambda) = A^\top (\beta(By - c) - \lambda)$. As shown in (Chan et al., 2024), it is sufficient to pass only (x, λ) between successive iterations.

Privacy setting. We consider an adversary that observes only (x, λ) at each step; y is recoverable from these variables.

Iteration invariant. We maintain $M^i x_i = x_0 + S^{(i)}$ for each batch index $i \geq 1$. The private procedure is detailed in Algorithm 6.

Synthetic Experiment: Setup and Additional Results. We run experiments on the generalized LASSO (elastic net) problem (Zou & Hastie, 2005). Given a dataset $U = \{(a_i, b_i) \in \mathbb{R}^n \times \mathbb{R}\}_{i \in [N]}$ and parameters c_1, c_2 , each user i is associated with $f_i(x) = (\langle a_i, x \rangle - b_i)^2$. For $x, y \in \mathbb{R}^n$, denote $F(x) = \frac{1}{N} \sum_{i \in [N]} f_i(x)$ and $g(y) = c_1 \|y\|_1 + c_2 \|y\|_2^2$, yielding $\min_{x, y} F(x) + g(y)$ subject to $x = y$.

Data generation. We follow (Goldstein et al., 2014) and generate synthetic data as:

1. For each $a_i \in \mathbb{R}^n$, set

$$a_{ij} = \begin{cases} 50z_{ij}, & \text{if } j \in [\lfloor n/5 \rfloor, \lfloor 2n/5 \rfloor], \\ z_{ij}, & \text{otherwise} \end{cases}$$

where $z_{ij} \sim \mathcal{N}(0, 1)$ independently.

Algorithm 6 Private ADMM for Edit-Style Neighboring User Streams

```

1: Input: Differential privacy parameters  $(\varepsilon, \delta)$ ; a user stream  $\{f_t : \mathbb{R}^n \rightarrow \mathbb{R} \mid t \geq 1\}$ ;  $M := \mathbb{I} + \eta\beta A^\top A$  and
    $E(y, \lambda) := A^\top (\beta(By - c) - \lambda)$ ; an initial point  $(x_0, \lambda_0)$ 
2: Output: The adversary observes some pair  $(x_i, \lambda_i)$  for each step  $t \geq 1$ , where  $i$  is the batch index
3: Instantiate private interactive mechanisms RandBin, for edit-style neighboring user streams, with bin sizes in  $[U..2U]$ 
4: Instantiate adaptive PrivStreamSum, for Hamming-style neighboring vector streams, with overall  $(\varepsilon, \delta)$ -DP guarantee,
   where  $U = O(\frac{1}{\varepsilon} \log \frac{1}{\delta})$ 
5:  $i \leftarrow 0$   $\triangleright$  batch index
6: for each time step  $t \geq 1$  at which user  $f_t$  arrives do
7:    $B_t \leftarrow \text{RandBin}(f_t)$   $\triangleright$  potential batch of released users
8:   if  $B_t = \perp$  then
9:     Output  $(x_i, \lambda_i)$ 
10:  else
11:     $i \leftarrow i + 1$ 
12:     $y_{i-1} \leftarrow \mathcal{G}(x_{i-1}, \lambda_{i-1})$ 
13:     $\lambda_i \leftarrow \lambda_{i-1} - \beta(Ax_{i-1} + By_{i-1} - c)$ 
14:     $d^{(i)} \leftarrow -\frac{\eta}{U} \cdot M^{i-1} \sum_{f \in B_t} (\nabla f(x_{i-1}) + E(y_{i-1}, \lambda_i))$   $\triangleright$  mini-batch gradient update
15:     $S^{(i)} \leftarrow \text{PrivStreamSum}(d^{(i)})$   $\triangleright$  private prefix sum of vectors so far
16:     $x_i \leftarrow M^{-i}(x_0 + S^{(i)})$ 
17:    Output  $(x_i, \lambda_i)$ 
18:  end if
19: end for

```

2. Let $x' \in \mathbb{R}^n$ have $x'_j = 3$ for $j \in [n/5, 2n/5]$ and 0 otherwise. Set $b_i = \langle a_i, x' \rangle + \xi_i$ where $\xi_i \sim \mathcal{N}(0, \sigma_b^2)$ independently.

Parameter sweeps. We fix $n = 64$, $N = 10000$, $\beta = 0.15$, and $\eta = 0.05$. We set $\sigma = 0.01, 0.1, 1$ and consider $\varepsilon \in \{0.5, 1, 2, 5\}$, $\delta \in \{10^{-5}, 10^{-6}\}$. We plot objective values against both the number of users and the number of returned batches.

Random vs. fixed batch sizes. We compare randomized batch sizes in $[U..2U]$ with a fixed batch size $\frac{3U}{2}$. Figure 9 shows nearly identical behavior; small privacy parameters can delay the first update due to larger U .

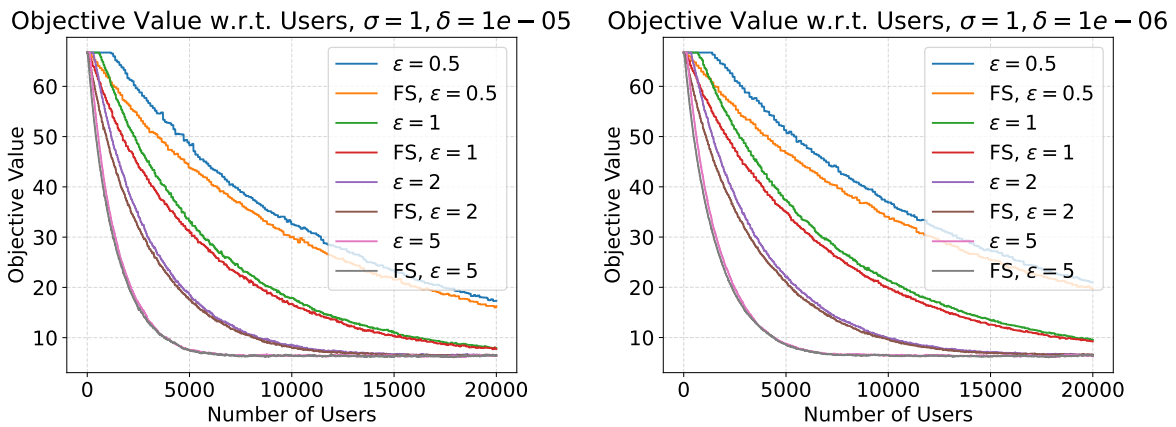


Figure 9. Each subfigure compares random versus fixed batch sizes for four values of ϵ as users arrive.

Batch-based views. Figure 10 plots objective values against the number of returned batches.

User-based views. We provide sweeps across (ε, δ) per noise level in Figure 11. The same qualitative trends persist: tighter privacy (smaller ε, δ) yields larger batches and slower updates, while batch-indexed views collapse more tightly

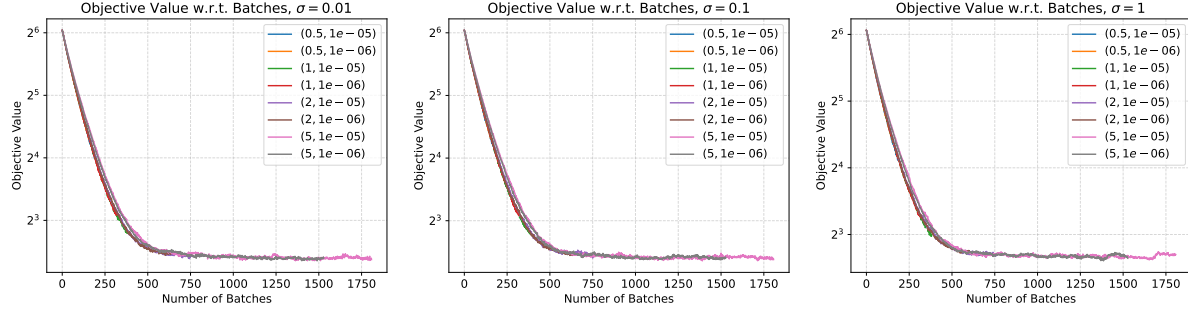


Figure 10. Each subfigure compares different ϵ as batches are returned.

across parameter choices.

- Smaller privacy parameters lead to larger batch sizes and less frequent model updates, so objective values converge slower. Towards the end, larger batches yield smoother curves due to more accurate gradient estimates.
- Combining with the batch-based views, convergence is mostly driven by the number of model updates. Smaller privacy parameters can balance larger aggregation noise with more accurate gradient estimates from larger batches, resulting in smoother curves.

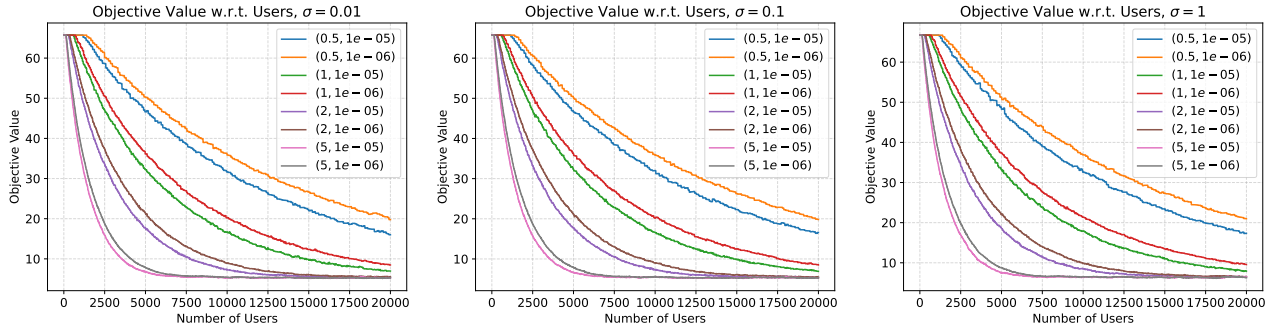


Figure 11. Each subfigure compares different ϵ and δ as users arrive.

H. Other Related Work

As the most relevant related works have already been covered in the introduction, we will elaborate further on some related aspects.

Choices of Privacy Measure. The central idea in differential privacy (Dwork, 2006) is that if V_0 and V_1 are two distributions of outputs produced by a mechanism from two neighboring inputs, then those two distributions should be close. The classical (ϵ, δ) -DP notion uses two parameters to quantify closeness, where a smaller value in each parameter means that the two distributions are closer. However, since distributions are inherently complex objects, some information on the two distributions will be inevitably lost when they are compared using just two parameters. When the same Gaussian noise is used to mask two different vectors, researchers have discovered that the Rényi divergence (Mironov, 2017) can capture the variance of the Gaussian distribution perfectly, and hence, can quantify the closeness of two such distributions (with the same variance but different means).

In general, any useful way to quantify privacy guarantees must satisfy the property that if the output satisfies certain privacy requirement, then any further processing of the output cannot violate that specific requirement. This can be formally formulated by requiring that the divergence – used for measuring how different two distributions are – must satisfy the *data processing inequality*; it is worth noting that some common distance notion such as the ℓ_2 -norm do not satisfy this property.

Instead of just using a few parameters to capture the closeness of two distributions, *tradeoff functions* (Dong et al., 2022)

have been proposed to define differential privacy, because a tradeoff function can capture all the essential information about how two distributions differ in the sense that any divergence satisfying the data processing inequality can be recovered from the tradeoff function. Indeed, tradeoff functions offer a powerful tool to describe the composition of private mechanisms.

However, one notational inconvenience is that a larger tradeoff (measured by pointwise comparison) means that the two distributions are closer, which has the opposite interpretation from other divergence parameters such as ε and δ . In fact, an *ad hoc* concept of *generalized probability distance* has been defined in (Vadhan & Zhang, 2023) to reverse the direction of the inequalities such that it will be consistent with the notion of distance. On the other hand, a simpler way to achieve this notation consistency is to replace a tradeoff function with its complement that is known as a *power function*, which naturally preserves all the equivalent mathematical properties. As we shall see in Definition A.3, a power function also has an intuitive description using the fractional knapsack problem.

As illustrated in (Vadhan & Zhang, 2023; Zhou et al., 2024), if one uses such a powerful tool to define any new notion of differential privacy, then a single composition theorem (such as our Theorem C.1) will be sufficient to recover any composition result from the classical notion to the new notion of privacy. Therefore, it would not be necessary to reconstruct individual advanced composition theorems (Dwork et al., 2010b; Kairouz et al., 2015).

Hamming vs Edit Neighboring Notions. In (Birrell et al., 2024), two neighboring notions are considered for **static** databases:

- *Hamming-style*. Two databases have the same number of elements, and they differ in at most one element.
- *Edit-style*. One element from one database is deleted to form the other database.

They considered fixed-size mini-batches, which may be sampled in two ways: with or without replacement. Since they considered sampling from static databases, the difference between Hamming- vs edit-style neighboring static databases would not have such a stark contrast as streams, as deleting the first element of a stream can cause it to change in every position.

Other Recent Works on Concurrent Composition. A more general notion of concurrent composition is considered in (Haney et al., 2023), where the privacy parameters of mechanisms can be adaptively chosen. However, as in (Vadhan & Zhang, 2023), each interactive mechanism is associated with a single static database on which its neighboring relation is defined; the adversary may interact adaptively with the mechanism, but neighboring inputs differ only in this underlying database, not in a dynamic stream of updates.

Concurrent composition for mechanisms with adaptively chosen privacy parameters is also considered in (Henzinger et al., 2026), but for neighboring dynamic databases. They also give a formulation based on adaptive DP (Denisov et al., 2022), expressed via a left-or-right style distinguishing game with a verification function, which is essentially the same as our *paired simulation* in Definition B.2.

Note that both works consider composition where each mechanism has a single neighboring relation on its dataset (static or dynamic). In order to capture modular composition in which neighboring notions are defined separately on both the input and the output of a mechanism—and to reason about neighbor-preserving transformations that change the neighboring structure—we need a more refined notion of *neighbor-preserving* paired simulation, given in Definition B.8.

I. Discussion, Limitations, and Takeaways

Interpretation: Privacy Implies Latency. In our setting, the buffering level U is *not* a tunable systems knob: it is fixed (up to constants) by the privacy allocation $(\varepsilon_b, \delta_b)$ via $U = \Theta(\varepsilon_b^{-1} \log(1/\delta_b))$. Consequently, the privacy target (ε, δ) induces a concrete batching and latency regime: stronger privacy (smaller ε and δ) forces larger bins and larger buffering backlog, which in turn increases delay. Our experiments therefore report utility versus (ε, δ) together with the empirical delay distributions implied by the corresponding $U(\varepsilon_b, \delta_b)$.

Relation to Convergent Privacy-Loss Analyses. Recent analyses of Noisy-SGD show that privacy loss can converge or saturate over long horizons (Altschuler & Talwar, 2022; Chien & Li, 2025). These results are complementary to our setting: they primarily analyze the privacy loss of optimizer dynamics, whereas Lemma 4.2 concerns the backlog and queueing delay induced by the edit-to-Hamming buffering wrapper. Extending such optimizer-specific analyses to jointly capture delayed incorporation and buffering-induced latency is an interesting direction for future work.

Pure-DP Scope. A positive δ in our (ε, δ) -DP construction is not merely a cosmetic convenience. The probability-one delay bound relies on bounded-support buffering noise and on an always-valid prefix-sum error bound; in our implementation, both are obtained in the approximate-DP regime. For a RandBin-style shift-masking first boundary, pure DP would require unbounded-support noise, so the same bounded-support, probability-one finite delay guarantee is impossible for this proof strategy.

This should not be read as a blanket impossibility theorem for all possible pure-DP edit-private continual mechanisms. If bounded support and probability-one finite delay are dropped, pure DP is not ruled out a priori. However, the present NPDP/NPP proof architecture faces an additional underflow obstacle: with unbounded bin-size or scheduler noise, the scheduler may request an emission before enough real items are available. Simple fixes such as padding or aborting do not preserve the current neighboring-bin alignment argument, since one execution may need to pad or abort exactly where the neighboring execution emits an ordinary bin. Thus our current pure-DP barrier is a limitation of the RandBin-style bounded-delay construction and proof architecture, rather than a general impossibility claim.

Limitations. First, our end-to-end guarantee is stated for *single-edit* neighboring user streams (one insertion/deletion). Handling multiple edits can be obtained via standard composition/group-privacy reasoning, but doing so necessarily weakens the privacy parameters and correspondingly increases the induced latency through $U(\varepsilon_b, \delta_b)$. Second, the certification theorem is intentionally conservative: it applies to quantitatively decomposable certified cores, combining prefix-causal fresh randomness with stability of the one-round privacy profile under common aggregation context. Mechanisms that reuse randomness across rounds, depend on future inputs, or allow common adaptive inputs to change the leakage profile fall outside Checklist 1 and require separate interaction-aware analysis. Third, we focus on a streaming SGD instantiation; other optimizers fit the same template, but we do not claim the pipeline resolves broader issues such as client clustering or personalization in federated settings.

Practical Mapping: Events, Releases, and Rounds. Delay is naturally measured in *events*: $D(i)$ counts subsequent arrivals until f_i is binned. If one prefers a “round” interpretation, the relevant unit is the emitted-bin index k (i.e., the number of releases), since the model is updated only at emission times; the event-level delay distribution then induces a distribution over the number of releases an event waits.

Takeaway. The main contribution is an *auditable* continual-learning recipe: randomized buffering reduces single-edit to a Hamming-style interface with explicit backlog guarantees, certification identifies when standard continual DP primitives remain private under adaptive interaction, and modular composition yields an end-to-end DP guarantee that is validated empirically together with its privacy-implied latency.